

The Kitaev Honeycomb Model: From Bogoliubov-de Gennes Transformations to the Quantum Spin Liquid Phase Diagram

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Abstract

These lecture notes focus on the Kitaev honeycomb model, a two-dimensional quantum system with three distinct Ising-type interactions, depending on the direction of each bond. This model has proven particularly important for describing quantum spin liquids, as well as for applications in topological quantum computing, due to its exotic topological properties. It constitutes the first exactly solvable theoretical model that fully captures the behavior of a quantum spin liquid. We begin by presenting the geometry of the model, the Hamiltonian that describes the system, and Alexei Kitaev's effort to obtain an exact analytical solution, based on the decomposition of the Hilbert space into flux sectors. We then present Ettore Majorana's observation that the Dirac equation also admits real solutions, leading to the theoretical prediction of Majorana fermions – particles that are identical to their antiparticles. We examine how the spatial separation of Majorana operators leads to the rise of unbound Majorana fermions, and showcase the exotic topological properties that manifest in physical systems such as superconducting wires. Next, we represent the Hamiltonian in the basis of Majorana fermions, by expressing spin operators using four Majorana operators. The transition from the extended space $\tilde{\mathcal{L}}$ to the physical subspace $\mathcal{L}$ is achieved through a gauge transformation of the $\mathbb{Z}_2$ group. We then reformulate the Hamiltonian in the basis of complex bond and matter fermions, transition to momentum space, and assume that the system lies in the zero-flux sector. Through the Bogoliubov–de Gennes transformation, we arrive at the final diagonalized Hamiltonian in the quasiparticle basis. Finally, we represent the possible phases – the gapless phase (phase B) and the gapped phases (phase A_{α_i}) – compute the corresponding energy spectrum, and present the complete phase diagram. By employing the correlation function, we show that all phases are characterized by strong quantum entanglement and spin fractionalization, classifying the system as a quantum spin liquid.

Contents

1	Honeycomb Kitaev Model	3
1.1	Lattice geometry	3
1.2	Hamiltonian of lattice	5
1.3	Flux operator $\hat{w}_{p_n}$	5
1.4	Hilbert space divided into sectors	7
2	Majorana Fermions	8
2.1	Dirac equation	8
2.2	Majorana fermions	10
2.3	Systems with a large number of electrons	11

2.4	1D Topological Superconducting Wire	12
2.5	Fermionic Fock space	13
3	Representation of Hamiltonian by Majorana operators	14
3.1	Representation of spins by Majorana operators	14
3.2	Multi spin systems	16
3.3	Hamiltonian in terms of majorana fermions	16
3.4	Complex bond and matter fermions (cbf) & (cmf)	19
3.5	Unbound Majorana modes (UMBs) & Majorana zero modes (MZMs)	21
4	Relation Between Hamiltonians and Bogoliubov Transformations	22
4.1	Hamiltonian matrix in terms of Majorana fermions	23
4.2	Hamiltonian matrix in terms of cbf and cmf	24
4.3	Bogoliubov-de Gennes (BdG) transformations	25
4.4	Ground State Flux Sector	28
5	Phase diagram and energy spectrum $E(q_x, q_y)$	31
5.1	Gapless phase	31
5.2	Gapped phase	33
5.3	Phase diagram	35
A	Example of sectors in Hilbert space	36
B	Pauli Matrices	36
C	Transformation of the flux operator $\hat{w}_{p_n}$	37
D	Proof of relationship $\tan 2\theta_q = -\frac{Im(S_q)}{Re(S_q)}$	38
E	Relationship between phase transitions and S_q	40
F	Correlation function $S_{\langle ij \rangle}^{\alpha\beta}(t)$	40
G	MATLAB Implementation for Computing the Energy Spectrum $E(q_x, q_y)$	43

List of Figures

1	Geometry of the honeycomb lattice in the Kitaev model. The x, y and z type bonds are indicated according to their orientation	4
2	The two alternating sublattices (odd and even) of the honeycomb lattice, on which the structure of the Kitaev model is based.	4
3	The three types of interactions (x -, y - and z -bonds) that appear in the honeycomb lattice of the Kitaev model.	4
4	Representation of a hexagonal plaquette of the honeycomb lattice.	5
5	Representation of a hexagonal plaquette of the honeycomb lattice, where the bonds are described through the K_{ij} formalism.	6
6	Graphical representation of the Hamiltonian, as given in Eq. (31). Each spin corresponds to four distinct dots, each of which corresponds to a Majorana operator.	17

7	Graphical representation of complex bond and complex matter fermions on the lattice	19
8	The unit cell of the honeycomb lattice, as used in the Kitaev model. The figure also shows the lattice vectors that determine the directions of the x , y , and z bonds.	29
9	The energy spectrum $E(q_x, q_y)$ is shown for the isotropic case $J_x = J_y = J_z = 1$. Six zero-energy points (Dirac points) are observed within the 1 st Brillouin Zone (BZ), where $E(q_x, q_y) = 0$, indicating that the spectrum is gapless. The dispersion near these points is linear, forming characteristic Dirac cones.	32
10	The energy spectrum $E(q_x, q_y)$ is shown for the boundary case $J_x = J_y = 1$ and $J_z = 1.9$. We observe that there are only four Dirac cones within the 1 st BZ, where $E(q_x, q_y)$ vanishes. The Dirac points have begun to merge.	33
11	The spectrum $E(q_x, q_y)$ is shown above for $J_x = J_y = 1$ and $J_z = 2.5$. The system exhibits an energy gap and is in phase A_z	35
12	Phase diagram of the Kitaev model in the parameter space J_x, J_y, J_z . The regions A_{α_i} exhibit an energy gap (gapped), while region B is gapless. A point inside the above triangle describes the relative values of J_x, J_y, J_z . The isotropic point appears at the center of the triangle.	36
13	Comparison of a plaquette before and after the application of the K_{ij} formalism.	38
14	It is shown how a spin decomposes into two static π -fluxes and one dynamic cmf. The state $ \psi\rangle$ is a zero-flux state. Suppose the state $ \psi\rangle$ is in the zero-flux sector. Applying the operator σ_i^z to the state $ \psi\rangle$, where site i is connected to site j through the bond $\langle ij \rangle$, yields the new state $ \psi'\rangle$. The state $ \psi'\rangle$ contains two static π -fluxes in the plaquettes between the bond $\langle ij \rangle$, as well as one dynamic cmf, represented as a black circle.	41
15	The above graph shows the correlation $\langle \sigma_{1z} \sigma_{2z} \rangle$ on a given z -bond. We have set $J_x = J_y = 0.5$ and vary J_z from 0 to 1. As expected, for $J_z = 0$ the correlation is zero, while for large values of J_z it converges to 1.	43

1 Honeycomb Kitaev Model

Summary

In Sec. 1, based on [1] and [2], we first analyze the geometry of the model. We then present the Hamiltonian of the system and, finally, Alexei Kitaev's effort to find an analytical solution by dividing the Hilbert space into sectors.

1.1 Lattice geometry

In the Honeycomb Kitaev model, a honeycomb lattice structure appears, where at each vertex of the honeycomb structure a particle with spin $\pm 1/2$ is hosted. The honeycomb lattice is divided into two equivalent sublattices (see Fig. 2). The sublattice called odd has a black dot as its center and three white dots as nearest neighbors, and the sublattice called even has a white dot as its center and three black dots as nearest neighbors. Depending on the direction in which the black dot is located relative to the white dot, three forms of interaction appear: x links, y links, z links (see Fig. 3).¹

¹: The black and white dots are not related to whether or not a particle is present. Depending on the geometric positions of the black dots relative to the white dots, they exhibit different forms of interaction

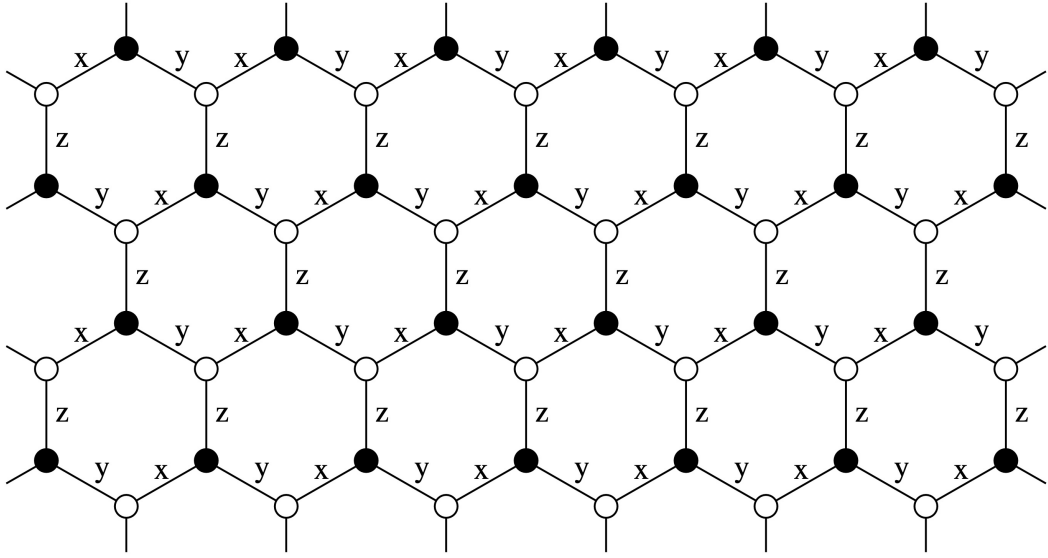


Figure 1: Geometry of the honeycomb lattice in the Kitaev model. The x , y and z type bonds are indicated according to their orientation

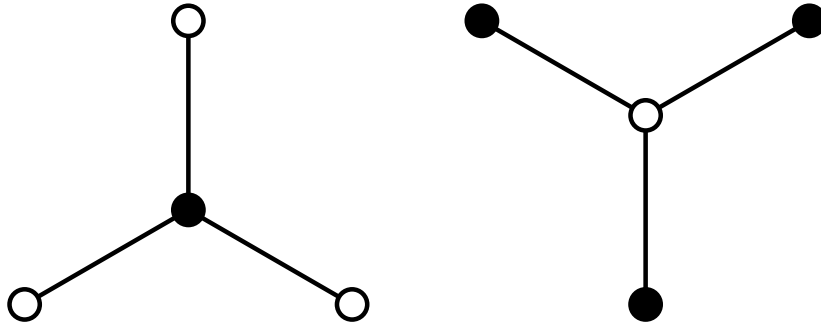


Figure 2: The two alternating sublattices (odd and even) of the honeycomb lattice, on which the structure of the Kitaev model is based.

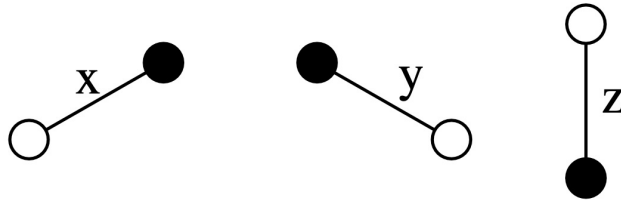
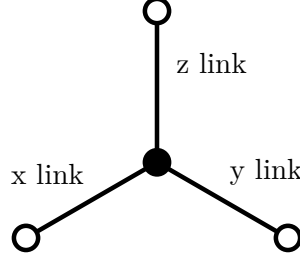


Figure 3: The three types of interactions (x -, y - and z -bonds) that appear in the honeycomb lattice of the Kitaev model.

(x -, y -, z links) between the particles.

1.2 Hamiltonian of lattice



To construct the expansion of the Hamiltonian we use only the odd sublattices. The expansion of the Hamiltonian for a random odd sublattice is:

$$\mathcal{H} = -J_x \sigma_i^x \sigma_j^x - J_y \sigma_i^y \sigma_j^y - J_z \sigma_i^z \sigma_j^z.$$

where $\sigma_i^{\alpha_i} \sigma_j^{\alpha_j}$ concerns nearest-neighbor interactions, with $\alpha_i = (x, y, z)$ depending on the type of bond (link).

For n odd sublattices we arrive at the Hamiltonian of the system:

$$\mathcal{H} = -J_x \sum_{\text{x-links}} \sigma_i^x \sigma_j^x - J_y \sum_{\text{y-links}} \sigma_i^y \sigma_j^y - J_z \sum_{\text{z-links}} \sigma_i^z \sigma_j^z, \quad (1)$$

where J_x , J_y , and J_z are coupling constants that describe the strength of the interaction along the x-, y- and z-bonds of the lattice. σ^x , σ^y and σ^z are the Pauli matrices². The Hamiltonian of the system exhibits a ferromagnetic-type coupling behavior. For each pair of the form $\sigma_i^{\alpha_i} \sigma_j^{\alpha_j}$, the spins tend to align in the same direction, i.e., either both are +1 or both are -1, so that the product is positive:

$$(+1)(+1) = +1, \quad (-1)(-1) = +1.$$

The sign in front of each coupling constant J_x , J_y and J_z is negative, in order to minimize the energy of the system when the spins are in a parallel arrangement.

1.3 Flux operator $\hat{w}_{p_n}$

Let us consider a random plaquette from the honeycomb lattice.

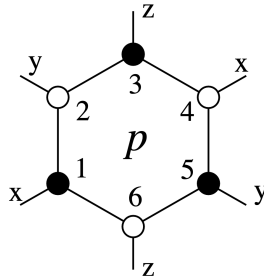


Figure 4: Representation of a hexagonal plaquette of the honeycomb lattice.

We define the flux operator:

²The form and derivation of the Pauli matrices can be found in Appendix A

$$\hat{w}_{p_n} = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z, \quad (2)$$

where p_n is the plaquette number.

Next, we introduce a new formalism for the Hamiltonian of the system, defining the operator K_{ij} :

$$K_{ij} = \begin{cases} \sigma_i^x \sigma_j^x, & \text{if } \langle i, j \rangle \text{ is an } x\text{-bond,} \\ \sigma_i^x \sigma_j^y, & \text{if } \langle i, j \rangle \text{ is a } y\text{-bond,} \\ \sigma_i^x \sigma_j^z, & \text{if } \langle i, j \rangle \text{ is a } z\text{-bond.} \end{cases} \quad (3)$$

Using the new formalism, the flux operator is:

$$\hat{w}_{p_n} = K_{12} K_{23} K_{34} K_{45} K_{56} K_{61}.^3 \quad (4)$$

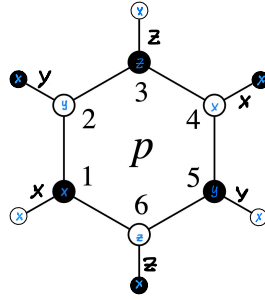


Figure 5: Representation of a hexagonal plaquette of the honeycomb lattice, where the bonds are described through the K_{ij} formalism.

We previously defined in Eq. (2) the flux operator:

$$\hat{w}_{p_n} = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z$$

Each operator $\sigma_i^{\alpha_i}$ acts exclusively on the corresponding particle i , without affecting the others. For example, σ_1^x acts only on particle 1, while σ_2^y acts only on particle 2, and so on. Each of these operators corresponds to a two-dimensional subspace within the total Hilbert space. Since the flux operator involves six particles, the full Hilbert space has dimension $2^6 = 64$. The matrix representation of $\hat{w}_{p_n}$ is:

$$\begin{aligned} \hat{w}_{p_n} = & (\sigma^x \otimes I \otimes I \otimes I \otimes I \otimes I) (I \otimes \sigma^y \otimes I \otimes I \otimes I \otimes I) \\ & (I \otimes I \otimes \sigma^z \otimes I \otimes I \otimes I) (I \otimes I \otimes I \otimes \sigma^x \otimes I \otimes I) \\ & (I \otimes I \otimes I \otimes I \otimes \sigma^y \otimes I) (I \otimes I \otimes I \otimes I \otimes I \otimes \sigma^z), \end{aligned} \quad (5)$$

where $\otimes$ corresponds to a direct sum of subspaces and I to the identity matrix. The first parenthesis corresponds to the representation of σ_1^x in the full Hilbert space. The same correspondence holds for $\sigma_2^y, \sigma_3^z, \sigma_4^x, \sigma_5^y$ and σ_6^z . Based on the mathematical property of the tensor product:

$$A \otimes B = (n \cdot p) \times (m \cdot q),$$

³The transformation of Eq. (2) to (4) is presented in detail in Appendix B.

where $A \in \mathbb{C}^{n \times m}$ and $B \in \mathbb{C}^{p \times q}$, it follows that the flux operator $\hat{w}_{p_n}$ is represented by a 64×64 matrix, since the total Hilbert space is $2^6 = 64$ dimensional. Such a large dimension makes the direct computation of eigenvalues of the matrix computationally difficult. However, we observe that the flux operator consists exclusively of products of Pauli matrices and identity matrices (see Eq. (5)). Since the eigenvalues of the Pauli matrices are ± 1 , while those of the identity matrix are 1, we conclude that the flux operator $\hat{w}_{p_n}$ will also exhibit eigenvalues ± 1 .

1.4 Hilbert space divided into sectors

Knowing now that the flux operator is defined as:

$$\hat{w}_{p_n} = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z = \prod_{\langle i,j \rangle} \sigma_i^{\alpha_i} \sigma_j^{\alpha_j} = K_{12} K_{23} K_{34} K_{45} K_{56} K_{61}. \quad (6)$$

We observe that the following quantities commute:

$$\begin{aligned} [\hat{K}_{ij}, \hat{w}_{p_n}] &= \hat{K}_{ij} \hat{w}_{p_n} - \hat{w}_{p_n} \hat{K}_{ij} = 0 \\ [\hat{w}_{p_i}, \hat{w}_{p_j}] &= \hat{w}_{p_i} \hat{w}_{p_j} - \hat{w}_{p_j} \hat{w}_{p_i} = 0 \\ [\hat{w}_{p_i}, \hat{\mathcal{H}}] &= \hat{w}_{p_i} \hat{\mathcal{H}} - \hat{\mathcal{H}} \hat{w}_{p_i} = 0 \end{aligned}$$

The Hamiltonian $\hat{\mathcal{H}}$ commutes with the flux operator $\hat{w}_{p_n}$, as a result the two quantities share a common system of eigenstates. Furthermore, the flux operators $\hat{w}_{p_i}$ and $\hat{w}_{p_j}$ commute with each other. This means that these operators can be measured simultaneously, allowing for a common eigenbasis. At this point, Kitaev's particularly clever idea emerges: to simplify the problem by dividing the system into discrete sectors of fixed flux, in which the values of $\hat{w}_{p_i}$ remain constant. In this way, the search for the eigenvalues of the Hamiltonian is restricted to each sector separately, drastically reducing the complexity of the problem.

The Hamiltonian of the system is now defined as:

$$\mathcal{H} = -J_{\alpha_i} \sum_i \hat{w}_{p_i}, \quad i : \text{plaquette number}. \quad (7)$$

We divide the space $\mathcal{L}$ of the system into a sum of flux sectors:

$$\mathcal{L} = \bigoplus_{w_{p_1}, \dots, w_{p_i}} \mathcal{L}_{w_{p_1}, \dots, w_{p_i}}, \quad (8)$$

Let us examine a specific flux sector:

$$\mathcal{L}_{w_{p_1}, \dots, w_{p_i}}.$$

Each unit cell in the lattice consists of 6 vertices. Each vertex contributes to a total of 3 unit cells, so that each plaquette consists of 2 vertices. We therefore arrive at the relation $m = \frac{n}{2}$, where m denotes the total number of plaquettes and n the number of vertices in the honeycomb lattice. The dimension of the sector is given by:

$$\frac{2^n}{2^m} = 2^{\frac{n}{2}}.$$

For $n = 100$, the sector⁴ has 2^{50} dimensions, an enormous number corresponding to an extremely large number of degrees of freedom; the goal of finding eigenvalues in each sector becomes practically impossible. We therefore conclude that the division of the Hilbert space into flux sectors, although it theoretically reduces complexity, is not sufficient for the analytical solution of the problem. In Sec. 4, we will present a new method that will allow the exact computation of the eigenvalues of the system.

2 Majorana Fermions

Summary

In Sec. 2, based on [3], [4], [5] and [6], we first present, through the Dirac equation, how Majorana fermions emerge. We then analyze the use of Majorana operators in systems with a large number of electrons and, finally, present physical systems in which unbound Majorana fermions appear. The analysis of Sec. 2 provides a basic mathematical background for better understanding of the following Sections.

2.1 Dirac equation

The Dirac equation is the fundamental equation for the relativistic description of spin-1/2 particles, incorporating special relativity into quantum mechanics. The structure of the equation allows for the prediction of the existence of antiparticles and the distinction between positive and negative energy states. In this Section we present the form of the Dirac equation and then investigate the conditions under which it can be reduced to real form, leading to the description of Majorana fermions — particles identical to their own antiparticles. We begin with the most general form of the Dirac equation⁵:

$$i\frac{\partial\Psi}{\partial t} = \hat{\mathcal{H}}_{\text{Dirac}}\Psi = (\alpha^i \cdot p + \beta m)\Psi, \quad (9)$$

where α^i ($i = x, y, z$) are matrices that describe the relativistic coupling of momentum to the spin of the particle. The matrix β describes the contribution of mass to the energy level, separating particles from antiparticles (positive and negative energy).

$$\alpha^i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \quad (10)$$

We define the tensor $\gamma^\mu = (\gamma^0, \gamma^i) = (\beta, \beta\alpha^i)$, where γ^0 corresponds to the temporal component of the tensor and γ^i corresponds to the spatial component of the tensor. Eq. (11) takes the following form:

$$i\gamma^0\frac{\partial\Psi}{\partial t} = \gamma^0(\gamma^i p_i + m)\Psi, \quad \rightarrow \hat{\mathcal{H}}_{\text{Dirac}} = \gamma^0(\gamma^i p_i + m). \quad (11)$$

Applying the operator $\hat{\mathcal{H}}_{\text{Dirac}}$ twice to Eq. (11) gives very interesting solutions. The eigenvalues of the solutions are found as follows:

⁴See the example in Appendix C for more details about the sector

⁵For simplicity, we set $\hbar = c = 1$.

$$\begin{aligned}
(\hat{\mathcal{H}}_{\text{Dirac}})^2 &= (\gamma^0(\gamma^i p_i + m))^2 \\
&= \gamma^0(\gamma^i p_i + m)\gamma^0(\gamma^i p_i + m) \\
&= \gamma^0\gamma^i p_i \gamma^0\gamma^j p_j + \gamma^0\gamma^i p_i \gamma^0 m + \gamma^0 m \gamma^0\gamma^i p_i + \gamma^0 m \gamma^0 m \\
&= \gamma^0[m\gamma^0(m + \gamma^i p_i) + \gamma^i p_i \gamma^0(m + \gamma^j p_j)] \\
&= \gamma^0[m^2\gamma^0 + m\gamma^0\gamma^i p_i + m\gamma^i p_i \gamma^0 + \gamma^i p_i \gamma^0\gamma^j p_j] \\
&= \gamma^0[m^2\gamma^0 - m\gamma^i\gamma^0 p_i + m\gamma^i\gamma^0 p_i + \gamma^i p_i \gamma^0\gamma^j p_j], \quad [\gamma^\mu, \gamma^\nu]_+ = 2\eta_{\mu\nu}I \text{ and } \gamma^0\gamma^i = -\gamma^i\gamma^0 \\
&= (\gamma^0)^2 m^2 + (-\gamma^0)^2 \gamma^i \gamma^j (p_i p_j), \quad (\gamma^0)^2 = 1 \text{ and } \gamma^i \gamma^j = \delta^{ij}I \\
&= m^2 + (\delta^{ij} p_i p_j), \quad \text{for } i = j \\
&= m^2 + p^2,
\end{aligned} \tag{12}$$

where $\eta_{\mu\nu}$ corresponds to the Minkowski metric.

Substituting Eq. (11) into Eq. (12) leads to the following equation:

$$\begin{aligned}
\left(i\frac{\partial\Psi}{\partial t}\right)^2 &= (m^2 + p^2)\Psi, \quad p = -i\nabla \\
\Rightarrow \left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2\right)\Psi &= 0, \quad (\text{Klein-Gordon equation}).
\end{aligned} \tag{13}$$

Thus, we prove that $(\hat{\mathcal{H}}_{\text{Dirac}})^2 = E^2$, with $E^2 = m^2 + p^2$, which is the relativistic energy-momentum relation. The solutions of the Dirac equation correspond to four-component spinor wavefunctions, i.e. $\Psi \in \mathbb{C}^4$. This four-dimensional space arises from the fact that the Dirac matrices γ^μ , which satisfy the Clifford algebra [7] in 3 + 1 dimensions, have a minimal representation of 4×4 . Consequently, Ψ must be a four-dimensional spinor for the equation to be consistent. Each component corresponds to a physical state — particle or antiparticle, with spin up or down.

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}, \quad \begin{array}{l} \psi_1 : \text{particle with positive energy and spin "up"}, \\ \psi_2 : \text{particle with positive energy and spin "down"}, \\ \psi_3 : \text{antiparticle with negative energy and spin "up"}, \\ \psi_4 : \text{antiparticle with negative energy and spin "down"}. \end{array} \tag{14}$$

The discovery of antiparticles such as the anti-electron (positron) occurred shortly after Dirac's publication [8], and his work was regarded as a prediction of their existence. Let us now write Eq. (11) in a slightly different form, multiplying both sides by γ^0 :

$$\begin{aligned}
\gamma^0 i \gamma^0 \frac{\partial\Psi}{\partial t} &= ((\gamma^0)^2 \gamma^i p_i + (\gamma^0)^2 m)\Psi \\
i \frac{\partial\Psi}{\partial t} - \gamma^i p_i \Psi - m\Psi &= 0 \\
(i\gamma^\mu \partial_\mu - m)\Psi &= 0, \quad p^\mu = (p^0, p^i) = (i\partial_0, -i\partial_i) \\
(\gamma^\mu p_\mu - m)\Psi &= 0,
\end{aligned} \tag{15}$$

where $\partial_0 \equiv \frac{\partial}{\partial t}$, $\partial_i \equiv \frac{\partial}{\partial x_i}$.

The matrices that satisfy Eq. (15) are called Dirac matrices and are defined:

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \hat{\sigma}^\mu & 0 \end{pmatrix}, \quad (16)$$

where $\sigma^\mu = (\sigma^0, -\sigma^i)$, $\hat{\sigma}^\mu = (\sigma^0, \sigma^i)$, σ^0 is the identity matrix and σ^i $i = (x, y, z)$ are the Pauli matrices.

2.2 Majorana fermions

Ettore Majorana observed that, for particular choices of the matrices α^i and β , Eq. (11) can take a real form. Specifically, it becomes possible to choose the wavefunction ψ to be real, i.e. $\psi = \psi^*$. This form of the equation is a variant of Eq. (15) and describes particles which are simultaneously their own antiparticles, i.e. they satisfy the relation $\psi = \psi^c$, where ψ^c is the charge conjugate of ψ .

$$\begin{aligned} \psi &= \psi^c, \quad \psi^c : \text{antiparticle wavefunction} \\ &= C(\bar{\psi})^T, \quad C : \text{charge conjugation matrix} \\ &= C(\psi^\dagger \gamma^0)^T, \quad \text{where } C(\gamma^\mu)^T C^{-1} = \gamma^\mu. \end{aligned} \quad (17)$$

For Majorana fermions we know that $C = \gamma^0 \gamma^2$. Substituting $C = \gamma^0 \gamma^2$ into Eq. (17) leads to the following form:

$$\begin{aligned} \psi &= C(\psi^\dagger \gamma^0)^T \\ &= \gamma^0 \gamma^2 (\psi^\dagger \gamma^0)^T \\ &= \gamma^0 \gamma^2 \gamma^0 (\psi^\dagger)^T \\ &= \gamma^0 \gamma^2 \gamma^0 \psi^* \\ &= -(\gamma^0)^2 \gamma^2 \psi^* \\ &= -\gamma^2 \psi^* \\ &= \psi^*. \end{aligned}$$

We have thus shown that our spinor equals its own conjugate, i.e. $\psi = \psi^c$, which holds exclusively for Majorana spinors. Consequently, the wavefunction can be chosen to be real. Majorana fermions are particles that are identical to their own antiparticles and cannot be distinguished from each other. The equality $\psi = \psi^c$ expresses precisely this physical property. As a result, the space of spinorial states becomes invariant under the action of the charge conjugation operator $\hat{C}$, i.e. the application of $\hat{C}$ to any component of the spinor space leaves the system unchanged.

The matrices γ^μ in the Majorana representation are denoted $\tilde{\gamma}^\mu$ and take the form:

$$\begin{aligned} \tilde{\gamma}^0 &= i \begin{pmatrix} 0 & -\sigma^1 \\ \sigma^1 & 0 \end{pmatrix}, \quad \tilde{\gamma}^1 = i \begin{pmatrix} 0 & \sigma^0 \\ \sigma^0 & 0 \end{pmatrix}, \\ \tilde{\gamma}^2 &= i \begin{pmatrix} \sigma^0 & 0 \\ 0 & -\sigma^0 \end{pmatrix}, \quad \tilde{\gamma}^3 = \begin{pmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{pmatrix}. \end{aligned} \quad (18)$$

The Dirac equation can be regarded as a system of four coupled equations, each corresponding to one of the components of the four-dimensional spinor Ψ . Introducing the Majorana formalism, the equation can be split into two independent real subsystems,

each involving two coupled equations for the corresponding real variables. The solution of one of these subsystems describes a neutral spin- $\frac{1}{2}$ fermion that is identical to its own antiparticle, in accordance with the condition $\psi = \psi^c$.

2.3 Systems with a large number of electrons

Systems with multiple electrons are described via Hamiltonians that include creation and annihilation operators, with linear or nonlinear terms depending on the type of interactions. A characteristic example is the Hubbard model.

The action of a creation operator $\alpha^\dagger$ on an empty quantum state⁶ $|n\rangle$ creates a new occupied quantum state⁷ $|n+1\rangle$

$$\alpha^\dagger |n\rangle = |n+1\rangle.$$

The action of a creation operator $\alpha^\dagger$ on an occupied quantum state $|n\rangle$ gives 0; two electrons are not allowed to occupy the same quantum state (Fermi-Dirac statistics).

$$\alpha^\dagger |n\rangle = 0.$$

The action of an annihilation operator α on an empty quantum state $|n\rangle$ gives 0:

$$\alpha |n\rangle = 0.$$

The action of an annihilation operator α on an occupied quantum state $|n\rangle$ gives a new quantum state $|n-1\rangle$, from which it removes an electron:

$$\alpha |n\rangle = |n-1\rangle.$$

The creation and annihilation operators satisfy the following anticommutation relations:

$$[\alpha_i^\dagger, \alpha_j^\dagger]_+ = [\alpha_i, \alpha_j]_+ = 0, \quad [\alpha_i^\dagger, \alpha_j]_+ = \delta_{ij}.$$

Without any loss of generality we can express the operators $\alpha^\dagger$ and α in the Majorana basis:

$$\alpha_j = \frac{1}{2}(\gamma_{j1} + i\gamma_{j2}), \quad \alpha_j^\dagger = \frac{1}{2}(\gamma_{j1} - i\gamma_{j2}). \quad (19)$$

The Majorana operators satisfy the following anticommutation relations:

$$[\gamma_i, \gamma_j]_+ = 2\delta_{ij}I, \quad \gamma_{i\alpha}^\dagger = \gamma_{i\alpha}. \quad (20)$$

Think of operator γ_1 as the “real” part of the electron and γ_2 as its “imaginary” part. The two operators are bound together, i.e. they reside at the same spatial position and cannot be separated locally. Together they fully encode the information about the quantum state of the electron, as seen from their composition into a creation operator (see Eq. (19)). From relation (20) we observe that the operators are real and anticommuting. The action of a single Majorana operator γ on a quantum state $|n\rangle$ is not sufficient to determine the presence or absence of an electron; the information is not fully encoded. This is precisely why γ_1 and γ_2 always appear in linear combination.

⁶empty quantum state: a state with no electron

⁷occupied quantum state: a state in which an electron is present

Whether or not an electron is present depends on the combination of the two Majorana operators, i.e. on the occupation number n :

$$n = \alpha^\dagger \alpha = \frac{1}{2} (1 - i\gamma_1 \gamma_2).$$

The occupation number n takes the value 0 if no electron is present and the value 1 if an electron is present. The combination $i\gamma_1 \gamma_2$ determines the state of the system and can take the values ± 1 , corresponding to the two possible values of n . The quantity $i\gamma_1 \gamma_2$ is non-Hermitian:

$$(i\gamma_1 \gamma_2)^\dagger = -i\gamma_2 \gamma_1 = -i\gamma_1 \gamma_2.$$

Our quantity does not commute, so simultaneous measurement via this combination is impossible. We have thus shown again that the only way to fully encode the electron is through a linear combination of γ_1 and γ_2 . Therefore any electron system can be expressed in the Majorana basis. In most cases, this transformation has no particular meaning and merely complicates the calculations. The operators γ_1 and γ_2 reside at the same spatial position and their combination corresponds to an electron creation operator. Consequently, it makes no sense to express the electron itself as two separate entities in the Majorana basis. However, in topological superconducting systems there is a separation of Majorana fermions, i.e. γ_1 and γ_2 now reside at different spatial positions, making the representation of the system in the Majorana basis essential, since no other basis can describe the degrees of freedom of the system. Unbound Majorana fermions exhibit remarkably interesting topological properties with extraordinary applications in pioneering technologies [9],[10].

2.4 1D Topological Superconducting Wire

The separation of Majorana fermions becomes more easily understood when we express Eq. (19) in the form:

$$\gamma_{j1} = \alpha_j^\dagger + \alpha_j, \quad \gamma_{j2} = i(\alpha_j^\dagger - \alpha_j). \quad (21)$$

Inside a topological superconducting wire, as we know, electrons form Cooper pairs which move through the material without resistance. Representing the electrons now in the Majorana basis, we observe that the Majorana operators in the edge Cooper pairs — i.e. those located near the ends of the wire — become separated, forming unbound Majorana fermions. At very low temperatures, inside a topological superconducting wire and with a combination of an external perturbation to the system, the probability of the presence of a single electron at one of the two ends of the wire appears. Through the quantum tunneling effect, the electron has a probability of “binding” with one of the separated Majorana fermions, e.g. γ_1 or γ_2 , creating a new binary quantum state (presence or absence of a “bound” electron) at the end. This quantum state can be regarded as a quantum bit (qubit), representing quantum information at that location. The quantum states at the ends of the system are protected from perturbations of the external environment. This is due to the fact that perturbation of the individual γ operators does not change the electron’s information, simply because by themselves they do not fully encode the electron’s information, with the consequence that the quantum state is protected. These exotic phenomena are at the forefront of modern research, with particular interest in applications for the implementation of topological quantum computers [11].

2.5 Fermionic Fock space

In topological fermionic quantum systems we do not use the Hilbert space but the fermionic Fock space. In the most general case, it does not constitute a subspace of the Hilbert space, but is constructed as a direct sum of all Hilbert spaces with different numbers of fermions, from 0 to infinity:

$$\mathcal{F} = \mathcal{F}_0 \oplus \mathcal{F}_1 \oplus \mathcal{F}_2 \oplus \dots \quad (22)$$

- $\mathcal{F}_0$: is the subspace with 0 fermions (vacuum state).
- $\mathcal{F}_1$: is the subspace with 1 fermion.
- $\mathcal{F}_2$: is the subspace with 2 fermions, and so on.

Equivalently, we can write:

$$\mathcal{F} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \dots, \quad (23)$$

where $\oplus$ corresponds to a direct sum of subspaces. Relations (22) and (23) are equivalent; we usually prefer the notation (22). The Fock space consists of a sum of subspaces $\mathcal{F}_n$, where n is the number of fermions. In each subspace $\mathcal{F}_n$ all possible combinations of quantum states corresponding to the n fermions are present. Let us look at some simple examples for a better understanding of the interpretation of the space. We assume that the system under study consists of three quantum states, $|\alpha\rangle$, $|\beta\rangle$ and $|\gamma\rangle$. For the subspace $\mathcal{F}_2$ we will have the following combinations of quantum states:

- Antisymmetric state of two fermions in states α and β :

$$|\psi_{\alpha\beta}\rangle = \frac{1}{\sqrt{2}} (|\alpha\rangle \otimes |\beta\rangle - |\beta\rangle \otimes |\alpha\rangle).$$

- Antisymmetric state of two fermions in states α and γ :

$$|\psi_{\alpha\gamma}\rangle = \frac{1}{\sqrt{2}} (|\alpha\rangle \otimes |\gamma\rangle - |\gamma\rangle \otimes |\alpha\rangle).$$

- Antisymmetric state of two fermions in states β and γ :

$$|\psi_{\beta\gamma}\rangle = \frac{1}{\sqrt{2}} (|\beta\rangle \otimes |\gamma\rangle - |\gamma\rangle \otimes |\beta\rangle).$$

In the example of the topological superconducting wire, we define the subspace $\mathcal{F}_0$ as the subspace with zero bound electrons. We assume that at the left end of the wire the unbound Majorana fermions γ_1 and γ_2 are located, while at the right end γ_3 and γ_4 . In the subspace $\mathcal{F}_1$, where there is exactly one bound electron, we can form the following creation and annihilation operators: At the left end:

$$\alpha_L = \frac{1}{2}(\gamma_1 + i\gamma_2), \quad \alpha_L^\dagger = \frac{1}{2}(\gamma_1 - i\gamma_2).$$

At the right end:

$$\alpha_R = \frac{1}{2}(\gamma_3 + i\gamma_4), \quad \alpha_R^\dagger = \frac{1}{2}(\gamma_3 - i\gamma_4).$$

The two possible occupied states in the subspace $\mathcal{F}_1$ are described by the following combinations:

1. Occupied state at the left end:

$$|\psi_L\rangle = \alpha_L^\dagger |0\rangle = |1_L\rangle.$$

2. Occupied state at the right end:

$$|\psi_R\rangle = \alpha_R^\dagger |0\rangle = |1_R\rangle.$$

3 Representation of Hamiltonian by Majorana operators

Summary

In Sec. 3, based on [1], [2], [12], [13], we present how the Hamiltonian of the system can be expressed in the Majorana basis, the introduction of bond and matter fermions, and the technique for transitioning from the extended space to the physical subspace.

3.1 Representation of spins by Majorana operators

As we know, a fermionic system can be fully described using creation $\alpha^\dagger$ and annihilation α operators. Without any loss of generality, we express the two operators in the Majorana basis:

$$\gamma_{j1} = \alpha_j^\dagger + \alpha_j, \quad \gamma_{j2} = i(\alpha_j^\dagger - \alpha_j).$$

The representation of spin in the Majorana basis is achieved through four Majorana operators⁸, namely γ^x , γ^y , γ^z and γ , which correspond to the notation γ_1 , γ_2 , γ_3 and γ_4 . The four Majorana operators create a four-dimensional extended Fock space. The physical space of the system consists of a two-dimensional Hilbert space corresponding to the quantum states with spin up $|\uparrow\rangle$ and down $|\downarrow\rangle$. The four Majorana operators create more degrees of freedom than those corresponding to the physical Hilbert space, so some degrees of freedom do not correspond to physical properties of the system. The restriction of the extended Fock space to the physical Hilbert space is achieved through the $\mathbb{Z}_2$ symmetry. We first define the gauge operator $\hat{D} = \gamma^x \gamma^y \gamma^z \gamma$ (Gauge Operator, GO). Using the GO, we define the physical space $\mathcal{M}$ as a subspace of the extended space $\tilde{\mathcal{M}}$, i.e. $\mathcal{M} \subseteq \tilde{\mathcal{M}}$, subject to the condition:

$$|\psi\rangle \in \mathcal{M} \quad \text{if and only if} \quad \hat{D}|\psi\rangle = |\psi\rangle, \quad \text{where} \quad \hat{D} = \gamma^x \gamma^y \gamma^z \gamma. \quad (24)$$

We can think of the GO as a “filter” or “guardian” which allows us to identify which quantum states of the extended space $\tilde{\mathcal{M}}$ correspond to quantum states of the physical subspace $\mathcal{M}$.

$$\begin{aligned} \hat{D}|\psi\rangle &= +|\psi\rangle, \quad D = +1 \rightarrow \text{physical subspace}, \\ \hat{D}|\psi\rangle &= -|\psi\rangle, \quad D = -1 \rightarrow \text{unphysical subspace}. \end{aligned}$$

When the action of the GO on the state $|\psi\rangle$ gives eigenvalue $+1$, then the state belongs to the physical subspace. When the eigenvalue is -1 , then the state $|\psi\rangle$ does not belong to the physical subspace. The GO can be regarded as a gauge transformation (GT) of

⁸The representation of the Pauli operator algebra in the extended space is achieved using four Majorana operators.

the $\mathbb{Z}_2$ group.
GT:

$$\begin{aligned}\gamma^x &\rightarrow -\gamma^x, \gamma^y \rightarrow -\gamma^y, \gamma^z \rightarrow -\gamma^z, \gamma \rightarrow -\gamma, \\ D &\rightarrow -D, \\ -D|\psi\rangle &= -|\psi\rangle \rightarrow D = +1.\end{aligned}$$

We observe that any sign change in the four Majorana operators does not affect the physics of the system. A gauge symmetry of the $\mathbb{Z}_2$ group thus appears, i.e. the freedom to reformulate the description of the system without altering the observable physical quantities.

The representations of the Pauli matrices in the extended space $\tilde{\mathcal{M}}$ are:

$$\begin{aligned}\tilde{\sigma}^x &= i\gamma^x\gamma, \\ \tilde{\sigma}^y &= i\gamma^y\gamma, \\ \tilde{\sigma}^z &= i\gamma^z\gamma.\end{aligned}\tag{25}$$

For the description of the system in the extended space to be consistent, the Pauli matrices $\tilde{\sigma}^x, \tilde{\sigma}^y, \tilde{\sigma}^z$ must satisfy two basic conditions. Under GT they must be represented by the Pauli matrices of the physical subspace $\mathcal{M}$ and must obey the Pauli matrix algebra within $\mathcal{M}$. The matrices $\tilde{\sigma}^{\alpha_i}$ (where $\alpha_i = x, y, z$) are Hermitian and unitary:

$$(\tilde{\sigma}^{\alpha_i})^\dagger = \tilde{\sigma}^{\alpha_i}, \quad (\tilde{\sigma}^{\alpha_i})^2 = 1.$$

The matrices $\tilde{\sigma}^{\alpha_i}$ commute with the GO:

$$[\tilde{\sigma}^{\alpha_i}, \hat{D}] = 0.$$

Therefore, the two quantities share common eigenstates. Consequently, the action of $\tilde{\sigma}^{\alpha_i}$ does not change the eigenstate of the GO, ensuring compatibility with the physical subspace.

Before proceeding to Section 3.2, let us verify that relation (25) satisfies the two conditions.

Condition (1):

$$\begin{aligned}\tilde{\sigma}^x \tilde{\sigma}^y \tilde{\sigma}^z &= i\gamma^x \gamma^y \gamma^z \gamma \\ &= i\hat{D}. \\ \tilde{\sigma}^x \tilde{\sigma}^y \tilde{\sigma}^z |\psi\rangle &= i\hat{D} |\psi\rangle, \quad D = +1 \rightarrow \text{Physical subspace} \\ &= \sigma^x \sigma^y \sigma^z |\psi\rangle, \quad \text{where } i = \sigma^x \sigma^y \sigma^z \rightarrow \text{Consistent with the physical subspace.}\end{aligned}$$

Condition (2):⁹

$$\begin{aligned}[\sigma_i, \sigma_j] &= 2i\varepsilon_{ijk}\sigma^k \rightarrow \text{Pauli algebra.} \\ [\sigma_j^x, \sigma_j^y] &= [-i\gamma_j^y \gamma_j^z, -i\gamma_j^z \gamma_j^x] \\ &= -i[\gamma_j^y \gamma_j^z, \gamma_j^z \gamma_j^x] \\ &= -i(\gamma_j^y \{\gamma_j^z, \gamma_j^z\} \gamma_j^x - \gamma_j^z \{\gamma_j^x, \gamma_j^y\} \gamma_j^z) \\ &= 2(-i\gamma_j^y \gamma_j^x) \\ &= 2i\sigma_j^z \rightarrow \text{Consistent with the Pauli algebra.}\end{aligned}$$

⁹In Condition (2) the identity $[AB, CD] = A\{B, C\}D - C\{D, A\}B$ was used along with Eq. (25).

3.2 Multi spin systems

For multi-spin systems we have:

$$\begin{aligned}\tilde{\sigma}_j^{\alpha_i} &:= i\gamma_j^{\alpha_i}\gamma_j, \quad j : \text{particle index.} \\ \hat{D}_j &:= \gamma_j^x \gamma_j^y \gamma_j^z \gamma_j.\end{aligned}\tag{26}$$

Based on definition (3.3), we can express the full physical space $\mathcal{L}$ as a subspace of the full extended space $\tilde{\mathcal{L}}$, i.e. $\mathcal{L} \subseteq \tilde{\mathcal{L}}$, subject to the condition:

$$|\psi\rangle \in \mathcal{L} \quad \text{if and only if} \quad \hat{D}_j |\psi\rangle = |\psi\rangle, \quad \text{for all } j.\tag{27}$$

Without any loss of generality we can express the spins $\sigma_j^{\alpha_i}$ of the physical subspace $\mathcal{L}$ as a product of the GO and the Pauli matrices of the extended space $\tilde{\mathcal{L}}$:

$$\sigma_j^{\alpha_i} = \hat{D}_j \tilde{\sigma}_j^{\alpha_i},\tag{28}$$

$$\sigma_j^{\alpha_i} = i\gamma_j^{\alpha_i}\gamma_j, \quad D = +1 \text{ physical subspace.}\tag{29}$$

We expand relation (3.5):

$$\begin{aligned}\sigma_j^{\alpha_i} &= \hat{D}_j \tilde{\sigma}_j^{\alpha_i} \\ &= (\gamma_j^x \gamma_j^y \gamma_j^z \gamma_j)(i\gamma_j^{\alpha_i}\gamma_j) \\ &= -i\gamma_j^x \gamma_j^y \gamma_j^z \gamma_j^{\alpha_i} (\gamma_j)^2 \\ &= -i\gamma_j^x \gamma_j^y \gamma_j^z \gamma_j^{\alpha_i}, \quad \text{where } (\alpha_i = x, y, z).\end{aligned}$$

Substituting α_i leads to the relation:

$$\sigma_j^x = -i\gamma_j^y \gamma_j^z, \quad \sigma_j^y = -i\gamma_j^z \gamma_j^x, \quad \sigma_j^z = -i\gamma_j^x \gamma_j^y.\tag{30}$$

The minus sign is introduced so that the Pauli matrix algebra is satisfied. We have thus managed to express the operators σ^x , σ^y and σ^z exclusively through the three Majorana operators γ^x , γ^y and γ^z . For any fermionic system, we can reformulate the Hamiltonian $\mathcal{H}\{\sigma_j^{\alpha_i}\}$ as:

$$\tilde{\mathcal{H}}\{\gamma_j^{\alpha_i}, \gamma_j\} = \tilde{\mathcal{H}}\{\sigma_j^{\alpha_i}\},$$

where the equality implies that the Hamiltonian of the extended space $\tilde{\mathcal{L}}$ is restricted to the physical subspace $\mathcal{L}$, provided that Eq. (22) is satisfied.

3.3 Hamiltonian in terms of majorana fermions

First, before substituting relation (30) into the Hamiltonian of the system, we represent the Majorana operators γ^x , γ^y , γ^z and γ as dots (the reason for this will become clear shortly):

$$\begin{array}{ccc} & \gamma^z & \\ & \bullet & \\ & \bullet \gamma & \\ \gamma^x \bullet & & \bullet \gamma^y \end{array}$$

We start from the Hamiltonian of the system, as given in Eq. (1):

$$\mathcal{H} = -J_x \sum_{\text{x-links}} \sigma_i^x \sigma_j^x - J_y \sum_{\text{y-links}} \sigma_i^y \sigma_j^y - J_z \sum_{\text{z-links}} \sigma_i^z \sigma_j^z.$$

Applying transformation (4), the Hamiltonian is written:

$$\begin{aligned} \mathcal{H} &= -J_{\alpha_k} \sum_{j,k} K_{jk} \\ &= -J_{\alpha_k} \sum_{j,k} \tilde{K}_{jk} \\ &= -J_{\alpha_i} \sum_{\langle j,k \rangle} \tilde{\sigma}_j^{\alpha_i} \tilde{\sigma}_k^{\alpha_i}, \quad \text{applying relation (29)} \\ &= -J_{\alpha_i} \sum_{\langle j,k \rangle} (i\gamma_j^{\alpha_i} \gamma_j)(i\gamma_k^{\alpha_i} \gamma_k) \\ &= iJ_{\alpha_i} \sum_{\langle j,k \rangle} i\gamma_j^{\alpha_i} \gamma_k^{\alpha_i} \gamma_j \gamma_k, \quad \hat{u}_{jk} = (i\gamma_j^{\alpha_{jk}} \gamma_k^{\alpha_{jk}}) \quad \alpha_{jk} = \alpha_i \\ \tilde{\mathcal{H}} &= \frac{i}{4} \sum_{\langle j,k \rangle} \hat{A}_{jk} \gamma_j \gamma_k, \quad A_{jk} = \begin{cases} 2J_{\alpha_{jk}} \hat{u}_{jk}, & \text{if } j \text{ and } k \text{ are connected,} \\ 0, & \text{anything else.} \end{cases} \end{aligned} \quad (31)$$

In constructing the sum in Eq. (26), each bond u_{jk} is counted twice due to the structure of the lattice, which is corrected by introducing the factor $\frac{1}{2}$. The matrix A_{jk} additionally includes a factor of 2, to account for the full energy contribution of each bond per plaquette, since each bond participates in two different plaquettes. Finally, the addition of the factor $\frac{1}{2}$ balances the factor 2 of the matrix A_{jk} , leading to an overall factor of $\frac{1}{4}$. The graphical representation of Hamiltonian (31) is presented in Fig. 6.

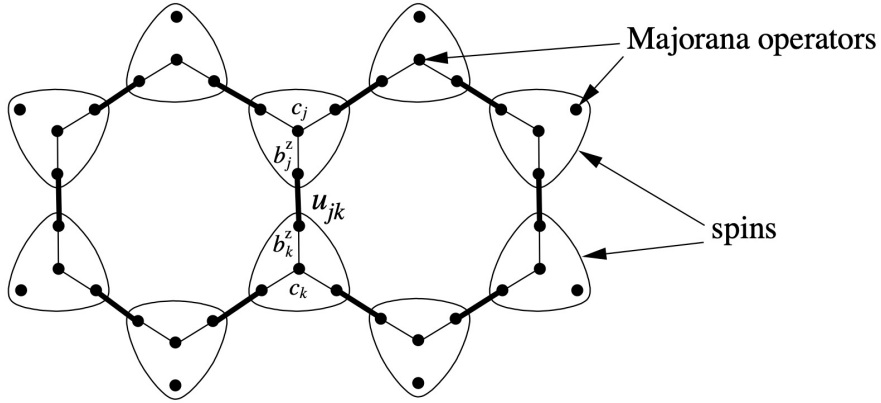


Figure 6: Graphical representation of the Hamiltonian, as given in Eq. (31). Each spin corresponds to four distinct dots, each of which corresponds to a Majorana operator.

The bond operator $\hat{u}_{jk}$ is antisymmetric, i.e. $u_{jk} = -u_{kj}$:

$$\begin{aligned}
\hat{u}_{jk}^\dagger &= (i\gamma_j^{\alpha_{jk}}\gamma_k^{\alpha_{jk}})^\dagger \\
&= ((i\gamma_j^{\alpha_{jk}}\gamma_k^{\alpha_{jk}})^T)^* \\
&= (i\gamma_k^{\alpha_{jk}}\gamma_j^{\alpha_{jk}})^* \\
&= -i\gamma_k^{\alpha_{jk}}\gamma_j^{\alpha_{jk}} \\
&= -u_{kj}, \quad \text{QED.}
\end{aligned}$$

The operator $\hat{u}_{jk}$ commutes with itself and with Eq. (31), $[\hat{u}_{jk}^i, \hat{u}_{jk}^s] = [\hat{u}_{jk}^i, \tilde{\mathcal{H}}] = 0$, and the two quantities share a common set of eigenstates. The eigenvalues of the operator $\hat{u}_{jk}$ are:

$$\begin{aligned}
u_{jk}^2 &= (i\gamma_j^{\alpha_{jk}}\gamma_k^{\alpha_{jk}})^2 \\
&= -(\gamma_j^{\alpha_{jk}}\gamma_k^{\alpha_{jk}})(\gamma_j^{\alpha_{jk}}\gamma_k^{\alpha_{jk}}) \\
&= \gamma_j^{\alpha_{jk}}\gamma_j^{\alpha_{jk}}\gamma_k^{\alpha_{jk}}\gamma_k^{\alpha_{jk}} \\
&= (\gamma_j^{\alpha_{jk}})^2(\gamma_k^{\alpha_{jk}})^2 \\
&= 1. \\
u_{jk} &= \pm 1.
\end{aligned}$$

Thus the only possible eigenvalues of the operator u_{jk} are ± 1 . We are now ready to decompose the extended space $\tilde{\mathcal{L}}$ of the system into sectors, where each sector corresponds to a possible combination of eigenvalues of the operator $\hat{u}_{jk}$:

$$\tilde{\mathcal{L}} = \bigoplus_u \tilde{\mathcal{L}}_u. \quad (32)$$

Eq. (31) is now written:

$$\tilde{\mathcal{H}} = \frac{i}{4} \sum_{\langle j,k \rangle} A_{jk} \gamma_j \gamma_k. \quad (33)$$

We remove the hat from the operator A_{jk} , since each sector is already separated into allowed combinations of eigenvalues of the operator $\hat{u}_{jk}$. Eq. (33) is a quadratic form.

The transition of the state $|\tilde{\Psi}_u\rangle$ from the extended space $\tilde{\mathcal{L}}$ to the physical subspace $\mathcal{L}$ is not as straightforward as in example at Eq. (27).

The extended space $\tilde{\mathcal{L}} = \bigoplus_u \tilde{\mathcal{L}}_u$ is not invariant under GT of the $\mathbb{Z}_2$ group.

The action of $\hat{D}_j$ on the bonds u_{jk} causes a reversal:

$$D_j u_{jk} D_j^{-1} = -u_{jk}, \quad \forall k \sim j. \quad (34)$$

More specifically, the action of $\hat{D}_j$ affects the bonds of vertex j that connect to the three nearest neighbors k_1 , k_2 and k_3 :

$$u_{jk_1} \rightarrow -u_{jk_1}, \quad u_{jk_2} \rightarrow -u_{jk_2}, \quad u_{jk_3} \rightarrow -u_{jk_3}.$$

Consequently we have a transition of the sector $\tilde{\mathcal{L}}_u$ to a new sector $\tilde{\mathcal{L}}'_u$:

$$\tilde{\mathcal{L}}_u \rightarrow \tilde{\mathcal{L}}'_u, \quad \text{where } u'_{jk} \rightarrow -u_{jk}, \quad \forall k \sim j. \quad (35)$$

As we know, the states $|\Psi_w\rangle$ in the physical subspace are gauge symmetric, i.e. invariant under GT. To achieve symmetrization of the state $|\tilde{\Psi}_u\rangle$ as it transitions to the physical subspace, we need to seek a mechanism that does not depend on the configurations of u_{jk} . The next section describes this mechanism in detail.

3.4 Complex bond and matter fermions (cbf) & (cmf)

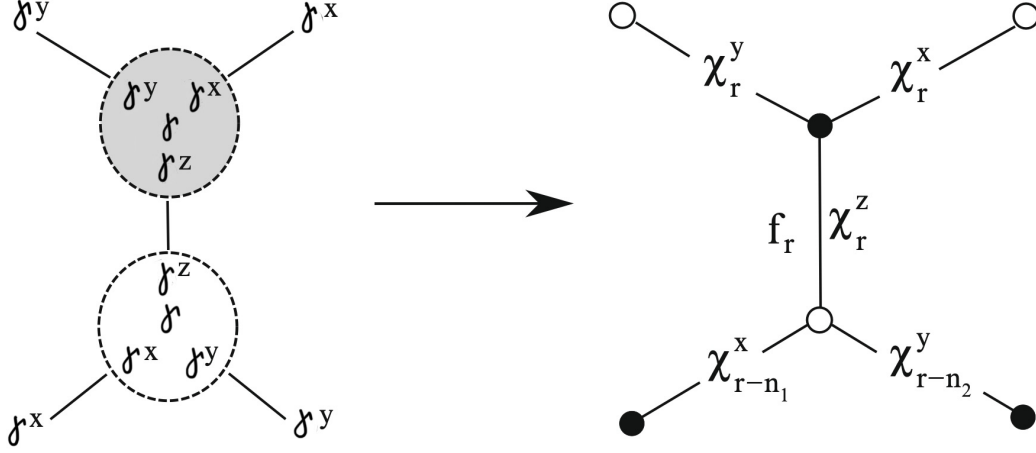


Figure 7: Graphical representation of complex bond and complex matter fermions on the lattice

We define the cbf through the operators $\chi_{\langle ij \rangle_\alpha}^\dagger$ and $\chi_{\langle ij \rangle_\alpha}$, which describe the state at the junctions of bond $\langle ij \rangle$, with the index α indicating the bond direction.

$$\chi_{\langle ij \rangle_\alpha} = \frac{1}{2} (\gamma_i^\alpha - i\gamma_j^\alpha), \quad \chi_{\langle ij \rangle_\alpha}^\dagger = \frac{1}{2} (\gamma_i^\alpha + i\gamma_j^\alpha). \quad (36)$$

You can think of the operators $\chi_{\langle ij \rangle_\alpha}^\dagger$ and $\chi_{\langle ij \rangle_\alpha}$ as creation and annihilation operators:

$$\begin{aligned} \chi_{\langle ij \rangle_\alpha}^\dagger &\rightarrow \text{creation of cbf at the bond,} \\ \chi_{\langle ij \rangle_\alpha} &\rightarrow \text{annihilation of cbf at the bond.} \end{aligned}$$

We transform Eq. (36) into the following form (to be used later):

$$\gamma_i^\alpha = \chi_{\langle ij \rangle_\alpha} + \chi_{\langle ij \rangle_\alpha}^\dagger, \quad i\gamma_j^\alpha = \chi_{\langle ij \rangle_\alpha} - \chi_{\langle ij \rangle_\alpha}^\dagger \quad (37)$$

The expression of the bond $u_{\langle ij \rangle_\alpha}$ in terms of $\chi_{\langle ij \rangle_\alpha}^\dagger$ and $\chi_{\langle ij \rangle_\alpha}$ is:

$$\begin{aligned} u_{\langle ij \rangle_\alpha} &= (i\gamma_i^\alpha \gamma_j^\alpha) \\ &= (\chi_{\langle ij \rangle_\alpha} + \chi_{\langle ij \rangle_\alpha}^\dagger)(\chi_{\langle ij \rangle_\alpha} - \chi_{\langle ij \rangle_\alpha}^\dagger) \\ &= \chi_{\langle ij \rangle_\alpha}^\dagger \chi_{\langle ij \rangle_\alpha} - (\chi_{\langle ij \rangle_\alpha}^\dagger)^2 + (\chi_{\langle ij \rangle_\alpha})^2 - \chi_{\langle ij \rangle_\alpha} \chi_{\langle ij \rangle_\alpha}^\dagger, \quad [\chi_{\langle ij \rangle_\alpha}^\dagger, \chi_{\langle ij \rangle_\alpha}]_+ = \delta_{ij} \\ &= 2\chi_{\langle ij \rangle_\alpha}^\dagger \chi_{\langle ij \rangle_\alpha} - 1. \end{aligned} \quad (38)$$

Depending on the value of $\chi_{\langle ij \rangle_\alpha}^\dagger \chi_{\langle ij \rangle_\alpha}$, the following results appear:

$$\begin{aligned} \chi_{\langle ij \rangle_\alpha}^\dagger \chi_{\langle ij \rangle_\alpha} &= +1, \quad \rightarrow u_{\langle ij \rangle_\alpha} = +1, \text{ Normal propagation.} \\ \chi_{\langle ij \rangle_\alpha}^\dagger \chi_{\langle ij \rangle_\alpha} &= 0, \quad \rightarrow u_{\langle ij \rangle_\alpha} = -1, \text{ Phase shift of } \pi, \text{ corresponding to a local flux.} \end{aligned}$$

We define the cmf through the operators f_r and $f_r^\dagger$, which describe the junctions of the central Majorana fermions from sublattice A (even) to sublattice B (odd):

$$f_r = \frac{1}{2}(\gamma_{A,r} + i\gamma_{B,r}), \quad f_r^\dagger = \frac{1}{2}(\gamma_{A,r} - i\gamma_{B,r}). \quad (39)$$

You can think of the operators $f_r^\dagger$ and f_r as creation and annihilation operators:

$$\begin{aligned} f_r^\dagger &\rightarrow \text{creation of cmf,} \\ f_r &\rightarrow \text{annihilation of cmf.} \end{aligned}$$

We transform Eq. (39) into the following form (to be used later):

$$\gamma_{A,r} = f_r + f_r^\dagger, \quad i\gamma_{B,r} = f_r - f_r^\dagger \quad (40)$$

Based on Eqs. (3.13) and (3.16), the gauge operator $\hat{D} = \gamma^x \gamma^y \gamma^z \gamma$ of sublattice A and B transforms as:¹⁰

$$D_{A,r} = [(\chi_r^x)^\dagger + \chi_r^x] [(\chi_r^y)^\dagger + \chi_r^y] [(\chi_r^z)^\dagger + \chi_r^z] [f_r^\dagger + f_r]. \quad (41)$$

$$D_{B,r} = [(\chi_{r-n_1}^x)^\dagger - \chi_{r-n_1}^x] [(\chi_{r-n_2}^y)^\dagger - \chi_{r-n_2}^y] [(\chi_r^z)^\dagger - \chi_r^z] [f_r^\dagger - f_r]. \quad (42)$$

Based on Eqs. (41) and (42), we define the projection operator as presented by Hong Yao in [13]:

$$P = \frac{1 + \sum_\nu D_\nu + \sum_{\nu < \nu'} D_\nu D_{\nu'} + \cdots + \prod_\nu D_\nu}{2^N}. \quad (43)$$

- The term 1: Includes the “vacuum state” (i.e. without any application of the D_ν).
- The term $\sum_\nu D_\nu$: Includes all individual local constraints.
- The term $\sum_{\nu < \nu'} D_\nu D_{\nu'}$: Includes all combinations of two local constraints.
- The term $\prod_\nu D_\nu$: Includes the full combination of all local constraints.
- The factor 2^N in the denominator is for normalization; also, the projection operator P must satisfy the condition $P^2 = P$.

Let us assume we are examining a system consisting of 8 Majorana fermions. This system includes $N = 2$ sites, i.e. $\nu = 1, 2$, with D_1 and D_2 corresponding to the local constraints. The projection operator, according to Eq. (43), is:

$$\begin{aligned} P &= \frac{1 + D_1 + D_2 + D_1 D_2}{4} \\ &= \prod_{\nu=1}^2 \frac{1 + D_\nu}{2} \quad (\text{Yao factorization}) \\ &= \frac{1 + D_1}{2} \cdot \frac{1 + D_2}{2} \\ &= \frac{1 + D_1 + D_2 + D_1 D_2}{4}. \end{aligned}$$

¹⁰Note that $\chi_{\langle ij \rangle \alpha} \equiv \chi_r^\alpha$ and $\chi_{\langle ij \rangle \alpha}^\dagger \equiv (\chi_r^\alpha)^\dagger$.

Thus Eq. (43) simplifies through the Yao factorization [13] as:

$$P = \prod_{\nu} \frac{1 + D_{\nu}}{2}. \quad (44)$$

Based now on the final form of the projection operator P , the transition of the state $|\tilde{\Psi}_u\rangle$ from the extended space $\tilde{\mathcal{L}}$ to the physical subspace $\mathcal{L} = \mathcal{L}_{\text{flux}} \otimes \mathcal{L}_{\text{matter}}$ is achieved:

$$|\Psi_w\rangle = \prod_{\nu} \frac{1 + D_{\nu}}{2} |\tilde{\Psi}_u\rangle. \quad (45)$$

Using the projection operator P , we have managed to symmetrize the quantum state $|\tilde{\Psi}_u\rangle$ with respect to all local GTs, i.e. we have managed to account for all possible configurations of u_{jk} arising from D_{ν} .

Example:

Suppose we have a configuration u_{jk} for a specific bond. As we know, the possible values of the bond are $u_{jk} = \pm 1$. The state $|\tilde{\Psi}_u\rangle$ depends on the configuration u_{jk} . Applying the projection operator P to $|\tilde{\Psi}_u\rangle$, we create a new state $|\Psi_w\rangle$ that includes both possible configurations; the new state no longer depends on the values u_{jk} due to its symmetrization:

$$|\Psi_w\rangle = \frac{1}{2}(|\tilde{\Psi}_{u_{jk}=+1}\rangle + |\tilde{\Psi}_{u_{jk}=-1}\rangle).$$

The states $|\Psi_w\rangle$ depend only on the topological class w_p :

$$w_{p_n} = \prod_{(j,k) \in \text{boundary}(p)} u_{jk}, \quad (j \in \text{even sublattice}, k \in \text{odd sublattice}). \quad (46)$$

The bonds (j, k) are taken with direction from vertex j , belonging to the even sublattice, to vertex k , belonging to the odd sublattice. This orientation ensures that w_{p_n} is computed consistently for all plaquettes. Eq. (46) consists of a product of six local bonds u_{jk} for each plaquette.

Example:

Suppose that for a specific plaquette the six local bonds have the following configuration $(u_1 = +1, u_2 = +1, u_3 = -1, u_4 = +1, u_5 = -1, u_6 = -1)$; the flux operator w_p then takes the following value:

$$\begin{aligned} w_p &= (u_1)(u_2)(u_3)(u_4)(u_5)(u_6) \\ &= (+1)(+1)(-1)(+1)(-1)(-1) \\ &= -1, \quad \text{flux present in the plaquette.} \end{aligned}$$

3.5 Unbound Majorana modes (UMBs) & Majorana zero modes (MZMs)

¹¹Suppose we are examining a specific plaquette of the system. As is known, each plaquette consists of six bonds (cbf) and three central matter fermion junctions (cmf) (see Fig. 6). The local flux $u_{\langle ij \rangle}$ at the bonds is related through Eq. (38), and can take the values $u_{\langle ij \rangle} = \pm 1$. When the product of the $u_{\langle ij \rangle}$ (see Eq. (46)) gives $w_p = +1$, the

¹¹The quantity w_p measures the local flux in a specific plaquette. The quantity W_p may refer to the overall state of the system or to a sum over plaquettes.

plaquette is in the flux-free phase. In this case, the cmf remain within their central positions and their wavefunction is considered delocalized. Very interesting phenomena appear when the plaquette is in the flux phase (vison), i.e. when $w_p = -1$. In this case, the wavefunctions of the cmf are considered localized, with the cmf remaining trapped within the boundaries of the plaquette and exhibiting exotic properties depending on the boundary conditions at the local bonds. If symmetry in the local bonds is observed around the cmf, i.e. $u_{\langle ij \rangle} = +1$ and $w_p = -1$, then the cmf remains localized within the boundaries of the plaquette and may exhibit pairing behavior similar to Cooper pairing. Conversely, in the case of symmetry breaking in the local bonds, i.e. $u_{\langle ij \rangle} = \pm 1$ and $w_p = -1$, the cmf splits into two UMBs¹², which move trapped within the boundaries of the plaquette.

Suppose the cmf is now located at the boundary between a flux-free plaquette ($w_p = +1$) and a plaquette with flux ($w_p = -1$). What will its behavior be? ¹³ We first define the energy cost of a flux change:

$$\Delta_{\text{flux}} := E_{\min}^{w_p=-1} - E_{\min}^{w_p=+1}. \quad (47)$$

If the hopping h is very small compared to Δ_{flux} , i.e. $h \ll \Delta_{\text{flux}}$, then the cmf cannot escape from the region with $w_p = -1$. In this case, it remains trapped in the flux region and splits into two UMBs. Conversely, when $h \gg \Delta_{\text{flux}}$, the cmf has enough kinetic energy to overcome the energy barrier of the flux region. It can then escape to regions with $w_p = +1$, where it moves freely.

The coupling energy between the UMBs is:

$$\Delta E \sim e^{-L/\xi}, \quad (48)$$

where ξ is the correlation length. In the case where the distance between the two UMBs becomes much larger than the correlation length, i.e. $L \gg \xi$, their coupling energy tends to zero, i.e. $\Delta E = 0$. This allows the formation of Majorana zero modes (MZMs). On 19/2/2025, Microsoft announced the Majorana 1 chip [14], which consists of six quantum qubits based on MZMs. This chip is based on Kitaev's model in one dimension [15] (Kitaev 1D chain). If the hopping h is comparable to the energy cost of a flux change, i.e. $h \sim \Delta_{\text{flux}}$, then the cmf may either become trapped or escape, depending on the local fluctuations. The probability of escape depends on the details of the gauge field and on thermal fluctuations of the system.

4 Relation Between Hamiltonians and Bogoliubov Transformations

Summary

In Sec. 4, based on [1], [2], [16] [17], [18], [19], [20], we present how the Hamiltonian transitions from the Majorana basis to the cbf and cmf basis. We then introduce the Bogoliubov-de Gennes transformations and present the diagonalized Hamiltonian in the ground state, which we use in Sec. 5 to derive the phase diagram.

¹²UMBs $\equiv$ Unbound Majorana fermions

¹³The terms Δ_{flux} and hopping h will be analyzed extensively in Sec. 4.

4.1 Hamiltonian matrix in terms of Majorana fermions

As we already know from Eq. (33), the Hamiltonian in the Majorana basis is defined:

$$\tilde{\mathcal{H}} = \frac{i}{4} \sum_{\langle j,k \rangle} A_{jk} \gamma_j \gamma_k, \quad A_{jk} = \begin{cases} 2J_{\alpha_{jk}} \hat{u}_{jk}, & \text{if } j \text{ and } k \text{ are connected,} \\ 0, & \text{anything else.} \end{cases}$$

We will attempt to express the Hamiltonian as a matrix product; later in the Section it will become clear why we chose this form.

We first express the Hamiltonian as a sum over sublattices A and B:

$$\tilde{\mathcal{H}} = \frac{i}{4} \sum_{\langle j,k \rangle} A_{jk} \gamma_{A_r} \gamma_{B_r}.$$

The two central Majorana fermions $\gamma_{A_r}, \gamma_{B_r}$ can be expressed via a vector:

$$\boldsymbol{\gamma} = \begin{pmatrix} \gamma_{A_r} \\ \gamma_{B_r} \end{pmatrix}.$$

We express A_{jk} as a (2×2) matrix consisting of four sub-matrices:

$$A = \begin{pmatrix} F & M \\ -M^T & -D \end{pmatrix}. \quad (49)$$

- F : describes interactions between two successive sublattices A (second neighbors).
- D : describes interactions between two successive sublattices B (second neighbors).
- M : describes interactions between sublattices A and B (first neighbors). The matrix M is defined as $M = u_{\langle ij \rangle} J_{\alpha_i}$.

In general, the matrix A represents the interactions between the various sites within the lattice. The matrix A is antisymmetric, hence ($F^T = -F$ and $D^T = -D$). The Hamiltonian of the system now takes the form:

$$\tilde{\mathcal{H}} = \frac{i}{2} \boldsymbol{\gamma}^T A \boldsymbol{\gamma}.$$

In matrix form:

$$\tilde{\mathcal{H}} = \frac{i}{2} \begin{pmatrix} \gamma_{A_r} & \gamma_{B_r} \end{pmatrix} \begin{pmatrix} F & M \\ -M^T & -D \end{pmatrix} \begin{pmatrix} \gamma_{A_r} \\ \gamma_{B_r} \end{pmatrix}. \quad (50)$$

In the Kitaev model we do not take into account interactions between second neighbors. We therefore assume that ($F = D = 0$), and the Hamiltonian becomes:

$$\tilde{\mathcal{H}} = \frac{i}{2} \begin{pmatrix} \gamma_{A_r} & \gamma_{B_r} \end{pmatrix} \begin{pmatrix} 0 & M \\ -M^T & 0 \end{pmatrix} \begin{pmatrix} \gamma_{A_r} \\ \gamma_{B_r} \end{pmatrix}. \quad (51)$$

Eq. (51) cannot capture the full picture in the Kitaev model. As we know, in the ground state we have absence of fluxes (no visons). All plaquettes exhibit local bonds $u_{\langle ij \rangle} = +1$, hence $w_{p_i} = +1$ for every i . In this case, due to translational symmetry¹⁴ of the system, we can assume that $F = D = 0$. The introduction of fluxes into the system causes breaking of translational symmetry, hence $F \neq D \neq 0$.

¹⁴Translational symmetry refers to the homogeneity of the system, i.e. whether the physical properties remain invariant under lattice translations

4.2 Hamiltonian matrix in terms of cbf and cmf

We are now ready to express Eq. (50) in the cbf and cmf basis.

Based on Eq. (40):

$$\gamma_{A_r} = f_r + f_r^\dagger, \quad \gamma_{B_r} = i(f_r^\dagger - f_r).$$

The form of Eq. (50) in the cbf and cmf basis is:

$$\tilde{\mathcal{H}} = \frac{i}{2} \begin{pmatrix} f_r + f_r^\dagger & i(f_r^\dagger - f_r) \end{pmatrix} \begin{pmatrix} F & M \\ -M^T & -D \end{pmatrix} \begin{pmatrix} f_r + f_r^\dagger \\ i(f_r^\dagger - f_r) \end{pmatrix}.$$

Step 1:

$$\begin{pmatrix} F & M \\ -M^T & -D \end{pmatrix} \begin{pmatrix} f_r + f_r^\dagger \\ i(f_r^\dagger - f_r) \end{pmatrix} = \begin{pmatrix} F(f_r + f_r^\dagger) + iM(f_r^\dagger - f_r) \\ -M^T(f_r + f_r^\dagger) - iD(f_r^\dagger - f_r) \end{pmatrix}.$$

Step 2:

$$\begin{pmatrix} f_r + f_r^\dagger & i(f_r^\dagger - f_r) \end{pmatrix} \begin{pmatrix} F(f_r + f_r^\dagger) + iM(f_r^\dagger - f_r) \\ -M^T(f_r + f_r^\dagger) - iD(f_r^\dagger - f_r) \end{pmatrix}.$$

From the multiplication in step 2 we arrive at the following Hamiltonian:

$$\begin{aligned} \tilde{\mathcal{H}} = \frac{i}{2} & \left[(f_r + f_r^\dagger)^T F (f_r + f_r^\dagger) + (f_r + f_r^\dagger)^T iM (f_r^\dagger - f_r) \right. \\ & \left. + i(f_r^\dagger - f_r)^T (-M^T) (f_r + f_r^\dagger) + i(f_r^\dagger - f_r)^T (-iD) (f_r^\dagger - f_r) \right]. \end{aligned}$$

We now simplify the four terms:

First term:

$$\begin{aligned} (f_r + f_r^\dagger)^T F (f_r + f_r^\dagger) &= \cancel{f_r^T F f_r} + f_r^T F f_r^\dagger + f_r^{\dagger T} F f_r + \cancel{f_r^{\dagger T} F f_r^\dagger} \\ &= f_r^T F f_r^\dagger + f_r^{\dagger T} F f_r. \end{aligned}$$

Second term:

$$(f_r + f_r^\dagger)^T iM (f_r^\dagger - f_r) = i f_r^T M f_r^\dagger - i f_r^T M f_r + i f_r^{\dagger T} M f_r^\dagger - i f_r^{\dagger T} M f_r.$$

Third term:

$$i(f_r^\dagger - f_r)^T (-M^T) (f_r + f_r^\dagger) = -i f_r^{\dagger T} M^T f_r - i f_r^{\dagger T} M^T f_r^\dagger + i f_r^T M^T f_r + i f_r^T M^T f_r^\dagger.$$

Fourth term:

$$\begin{aligned} i(f_r^\dagger - f_r)^T (-iD) (f_r^\dagger - f_r) &= \cancel{f_r^{\dagger T} D f_r^\dagger} - f_r^{\dagger T} D f_r - f_r^T D f_r^\dagger + \cancel{f_r^T D f_r} \\ &= -f_r^{\dagger T} D f_r - f_r^T D f_r^\dagger. \end{aligned}$$

The Hamiltonian takes the following form:

$$\begin{aligned} \tilde{\mathcal{H}} = \frac{i}{2} & \left[f_r^T F f_r^\dagger + f_r^{\dagger T} F f_r + i f_r^T M f_r^\dagger - i f_r^T M f_r + i f_r^{\dagger T} M f_r^\dagger - i f_r^{\dagger T} M f_r \right. \\ & \left. - i f_r^{\dagger T} M^T f_r - i f_r^{\dagger T} M^T f_r^\dagger + i f_r^T M^T f_r + i f_r^T M^T f_r^\dagger - f_r^{\dagger T} D f_r - f_r^T D f_r^\dagger \right]. \end{aligned}$$

We group the terms:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left[f_r^\dagger (M + M^T + i(F - D)) f_r + f_r^\dagger (M + M^T + i(F + D)) f_r^\dagger + f_r (M + M^T - i(F + D)) f_r \right].$$

We define the matrix elements:

$$\Delta = M - M^T + i(F + D), \quad (52)$$

$$h = M + M^T + i(F - D). \quad (53)$$

The Hamiltonian now in matrix form takes the form:

$$\tilde{\mathcal{H}} = \frac{1}{2} \begin{pmatrix} f_r^\dagger & f_r \end{pmatrix} \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^T \end{pmatrix} \begin{pmatrix} f_r \\ f_r^\dagger \end{pmatrix}. \quad (54)$$

- h : Corresponds to the hopping of cmf under a fixed gauge field background $u_{\langle ij \rangle}$. In simple terms, it expresses the probability of a cmf moving from site i to site j .
- Δ : Corresponds to the pairing of cmf and is related to topological superconductivity. That is, the cmf mimic the properties of Cooper pairs, with the only difference being that the pairing phenomenon is caused by the gauge field and the topology of the lattice, rather than by phonons as in the classical BCS model (Bardeen–Cooper–Schrieffer theory) [21].

4.3 Bogoliubov-de Gennes (BdG) transformations

We observe that the Hamiltonian of Eq. (54) has the BdG form. In general, Hamiltonians of BdG form describe superconducting systems. In the Kitaev model, the BdG form is used to explain the topological phases, where the cmf exhibit hopping terms and pairing terms.

We first apply the BdG transformation as presented in [20]. The importance of the BdG transformation will become apparent shortly:

$$\begin{pmatrix} f_r \\ f_r^\dagger \end{pmatrix} = T \begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix}. \quad (55)$$

The BdG transformation allows the expression of the cmf in a new basis consisting of quasiparticles (qsp) α_n . This transition is performed through a unitary operator $\hat{T}$, which satisfies:

$$T T^\dagger = 1.$$

The use of the operator $\hat{T}$ ensures that the physics of the system remains invariant, preserving the anticommutation properties of the fermionic operators, i.e.:

$$[\alpha_i, \alpha_j^\dagger]_+ = \delta_{ij}.$$

We define $\mathcal{H}_{BdG}$ as:

$$\mathcal{H}_{BdG} = \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^T \end{pmatrix}. \quad (56)$$

The eigenvectors and eigenvalues of $\mathcal{H}_{BdG}$ are found via the equation:

$$\mathcal{H}_{BdG} \Psi_n = E_n \Psi_n. \quad (57)$$

The eigenvectors of Eq. (57) have the form:

$$\Psi_n = \begin{pmatrix} X_n \\ Y_n \end{pmatrix}. \quad (58)$$

Substituting Eqs. (56) and (58) into (57) leads to the equations:

$$hX_n + \Delta Y_n = E_n X_n, \quad (59)$$

$$\Delta^\dagger X_n - h^T Y_n = E_n Y_n. \quad (60)$$

Based on Eqs. (59) and (60), and given that $TT^\dagger = 1$, the matrix T is obtained as a combination of the eigenvectors of $\mathcal{H}_{BdG}$:

$$T = \begin{pmatrix} X^T & Y^\dagger \\ Y^T & X^\dagger \end{pmatrix}. \quad (61)$$

The matrices X and Y have the same dimension as the Hamiltonian $\mathcal{H}_{BdG}$ and consist of a combination of the eigenvectors of $\mathcal{H}_{BdG}$. The matrices X and Y satisfy the following properties ensuring that the matrix T is unitary:

$$\begin{aligned} XX^\dagger + YY^\dagger &= I, \\ XY^T - YX^T &= 0. \end{aligned}$$

Knowing now the form of the matrix T , we are able, through Eq. (55), to present how the cmf operators are expressed in the basis of α_n . The α_n are a composite state that depends on both the local flux $u_{\langle ij \rangle}$ and the cmf:

$$\begin{aligned} f_{r_i} &= X_{ik}^T a_k + Y_{ik}^\dagger a_k^\dagger, \\ f_{r_j}^\dagger &= Y_{jl}^T a_l + X_{jl}^\dagger a_l^\dagger. \end{aligned}$$

The unitary transformation $T^\dagger \mathcal{H}_{BdG} T$ yields a diagonal matrix with the eigenvalues of $\mathcal{H}_{BdG}$ as its matrix elements. This is because the representation of the matrix T is in the eigenbasis of $\mathcal{H}_{BdG}$.

$$T^\dagger \mathcal{H}_{BdG} T = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}. \quad (62)$$

Applying the BdG transformation (see Eq. (55)) to Eq. (54) leads to the following form:

$$\begin{aligned}
\tilde{\mathcal{H}} &= \frac{1}{2} \begin{pmatrix} \alpha^\dagger & \alpha \end{pmatrix} T^\dagger \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^T \end{pmatrix} T \begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} \alpha^\dagger & \alpha \end{pmatrix} T^\dagger \mathcal{H}_{BdG} T \begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} \alpha^\dagger & \alpha \end{pmatrix} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \begin{pmatrix} \alpha \\ \alpha^\dagger \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} \alpha^\dagger & \alpha \end{pmatrix} \begin{pmatrix} E\alpha \\ -E\alpha^\dagger \end{pmatrix} \\
&= \frac{1}{2} \sum_{n>0} (\alpha_n^\dagger E_n \alpha_n - \alpha_n E_n \alpha_n^\dagger) \\
&= \frac{1}{2} \sum_{n>0} (E_n \alpha_n^\dagger \alpha_n - E_n \alpha_n \alpha_n^\dagger), \quad [\alpha_n^\dagger, \alpha_n]_+ = \delta_{ij} \\
&= \frac{1}{2} \sum_{n>0} (E_n \alpha_n^\dagger \alpha_n - E_n (1 - \alpha_n^\dagger \alpha_n)) \\
&= \sum_{n>0} E_n \alpha_n^\dagger \alpha_n - \frac{1}{2} \sum_{n>0} E_n \\
&= \sum_{n>0} E_n (\alpha_n^\dagger \alpha_n - \frac{1}{2}), \quad n : \text{the plaquette index.} \tag{63}
\end{aligned}$$

The excitations α_n are given by the eigenvalues E_n , which determine the behavior of the cmf under the effect of the flux.

We define the ground state as:

$$\alpha_i |0\rangle = 0, \tag{64}$$

i.e. the state of zero excitability of the system. This is the state in which there are no qsp, as they are locked in their positions under a fixed gauge field, i.e. in the flux-free sector. All qsp are in the ground level $E_{g.s}$ (vacuum state).

The energy of the ground state:

$$E_{g.s} = \langle 0 | \tilde{\mathcal{H}} | 0 \rangle = -\frac{1}{2} \sum_n E_n. \tag{65}$$

The first term in Eq. (63) describes the excitations of qsp moving in the gauge field. The change $w_p = +1 \rightarrow w_p = -1$ creates (excites) two free qsp to the higher energy level.

The flux change affects $E_{g.s}$ due to the variation of the local field $u_{\langle ij \rangle}$.

In Eq. (63) we did not address the distinction between the presence or absence of flux.

We will now present how the eigenstates of different flux sectors can be related to each other. Let β and α be the operators in which two different flux sectors are diagonalized.

With index F we denote a system with fluxes and with index 0 a system without fluxes.

Each system is diagonalized through its own BdG transformation [20]:

$$\begin{aligned}
\begin{pmatrix} X_0^* & Y_0^* \\ Y_0 & X_0 \end{pmatrix} \begin{pmatrix} f \\ f^\dagger \end{pmatrix} &= \begin{pmatrix} a \\ a^\dagger \end{pmatrix} \quad \text{inverse} \quad \begin{pmatrix} X_0^T & Y_0^\dagger \\ Y_0^T & X_0^\dagger \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix} = \begin{pmatrix} f \\ f^\dagger \end{pmatrix}, \\
\begin{pmatrix} X_F^* & Y_F^* \\ Y_F & X_F \end{pmatrix} \begin{pmatrix} f \\ f^\dagger \end{pmatrix} &= \begin{pmatrix} b \\ b^\dagger \end{pmatrix} \quad \text{inverse} \quad \begin{pmatrix} X_F^T & Y_F^\dagger \\ Y_F^T & X_F^\dagger \end{pmatrix} \begin{pmatrix} b \\ b^\dagger \end{pmatrix} = \begin{pmatrix} f \\ f^\dagger \end{pmatrix}.
\end{aligned}$$

We equate the two expressions:

$$\begin{pmatrix} X_0^T & Y_0^\dagger \\ Y_0^T & X_0^\dagger \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix} = \begin{pmatrix} X_F^T & Y_F^\dagger \\ Y_F^T & X_F^\dagger \end{pmatrix} \begin{pmatrix} b \\ b^\dagger \end{pmatrix}.$$

We multiply the equation by the inverse of $\begin{pmatrix} X_F^T & Y_F^\dagger \\ Y_F^T & X_F^\dagger \end{pmatrix}$:

$$\begin{aligned} \begin{pmatrix} X_F^* & Y_F^* \\ Y_F & X_F \end{pmatrix} \begin{pmatrix} X_0^T & Y_0^\dagger \\ Y_0^T & X_0^\dagger \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix} &= \begin{pmatrix} b \\ b^\dagger \end{pmatrix}, \\ \begin{pmatrix} X_F^* X_0^T + Y_F^* Y_0^T & X_F^* Y_0^\dagger + Y_F^* X_0^\dagger \\ Y_F X_0^T + X_F Y_0^T & Y_F Y_0^\dagger + X_F X_0^\dagger \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix} &= \begin{pmatrix} b \\ b^\dagger \end{pmatrix}, \\ \begin{pmatrix} \mathcal{X}^* & \mathcal{Y}^* \\ \mathcal{Y} & \mathcal{X} \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix} &= \begin{pmatrix} b \\ b^\dagger \end{pmatrix}. \end{aligned} \tag{66}$$

Relation (4.18) describes the transformation of qsp upon a change of the flux sector of the system. Essentially, it shows how the presence or absence of flux affects and transforms the qsp. This is important because it shows how the system transitions from one flux sector to another while preserving its topological structure. The study of the phase diagram in systems where fluxes are present is particularly difficult. The number of possible flux sectors grows exponentially with the size of the system. For N plaquettes, there are 2^N different flux states, making complete mapping of all configurations at large scales impossible. The plaquettes w_{p_i} are not independent, but are connected through the bonds $u_{\langle ij \rangle}$. If the flux in one plaquette changes, it can affect the neighboring fluxes, altering the overall energy configuration of the system. This creates macroscopic correlations that make the analytical study of the system more difficult.

4.4 Ground State Flux Sector

Suppose we wish to find the absolute minimum of the system; to achieve this we would need to compare all values of $E_{g,s}$ for all possible flux sectors. This is practically impossible. Fortunately, due to the translational symmetry of the system, we know that the ground state appears in the flux-free sector, i.e. when all plaquettes have eigenvalue $w_{p_i} = +1$. Although all plaquettes have the same eigenvalue $+1$, the configuration of the local bonds for each w_{p_i} is not necessarily the same (see Eq. (3.25)); hence the values in the sum $E_{g,s}$ (see Eq. (65)) may differ for each plaquette. We assume the system is at low temperatures, where all plaquettes have the same local configuration $u_{\langle ij \rangle} = +1$. The flux-free state is denoted $|\mathcal{G}^0\rangle$, with the corresponding ground state of the cmf denoted $|\mathcal{M}_0\rangle$. The probability that a bond $u_{\langle ij \rangle}$ changes follows the Boltzmann distribution:

$$P \sim e^{-\Delta_{\text{flux}}/k_B T}. \tag{67}$$

For low temperatures ($k_B T \ll \Delta_{\text{flux}}$), the probability of a bond changing is exponentially small.

We assume the system is at low temperatures, within the flux-free sector, i.e. in the ground state. For simplification of the calculations, we move to momentum space by applying a Fourier transform (Ft) to the cmf:

$$f_q = \frac{1}{\sqrt{N}} \sum_r e^{i\mathbf{q}\cdot\mathbf{r}} f_r,$$

With this transformation, Eq. (54) takes the following form in momentum space:

$$\tilde{\mathcal{H}}_0 = \begin{pmatrix} f_q^\dagger & f_{-q} \end{pmatrix} \begin{pmatrix} h_q & -\Delta_q \\ -\Delta_q^* & -h_q \end{pmatrix} \begin{pmatrix} f_q \\ f_{-q}^\dagger \end{pmatrix}. \quad (68)$$

We express the interaction potentials J_x, J_y, J_z in momentum space via the equation:

$$S_q = \sum_{n_i} J_{\alpha_{n_i}} e^{iqn_i}, \quad (69)$$

where the vectors $\mathbf{n}_1 = \mathbf{n}_x = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, $\mathbf{n}_2 = \mathbf{n}_y = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, $\mathbf{n}_3 = \mathbf{n}_z = (0, 0)$ are expressed in units of the bond length and describe the positions of neighboring atoms in the honeycomb lattice.

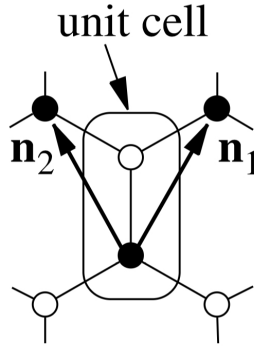


Figure 8: The unit cell of the honeycomb lattice, as used in the Kitaev model. The figure also shows the lattice vectors that determine the directions of the x , y , and z bonds.

We separate Eq. (69) into real and imaginary parts, $S_q = \text{Re}(S_q) + i\text{Im}(S_q)$:

$$\begin{aligned} S_q &= J_x e^{iqn_x} + J_y e^{iqn_y} + J_z \\ &= [J_x \cos(qn_x) + J_y \cos(qn_y) + J_z] + i[J_x \sin(qn_x) + J_y \sin(qn_y)] \\ \text{Re}(S_q) &= [J_x \cos(qn_x) + J_y \cos(qn_y) + J_z] \\ \text{Im}(S_q) &= [J_x \sin(qn_x) + J_y \sin(qn_y)]. \end{aligned}$$

In general, we know that in Kitaev models phenomena of p -wave superconductivity [22] (p -wave, $p-w$) appear. To ensure $p-w$, we require the pairing function to be purely imaginary and antisymmetric, i.e. $\Delta_q = -\Delta_{-q}$. Furthermore, the function h_q , which describes the dispersion of the cmf, i.e. how the energy of the cmf varies as a function of momentum, is required to be symmetric $h_q = h_{-q}$ and real. Consequently, the functions take the following form: $\Delta_q = -i\text{Im}(S_q)$ and $h_q = \text{Re}(S_q)$.

We define $\mathcal{H}_{BdG}$ in momentum space as:

$$\mathcal{H}_{BdG} = \begin{pmatrix} h_q & -\Delta_q \\ -\Delta_q^* & -h_q \end{pmatrix}.$$

Knowing now the form of Δ_q and h_q , the $\mathcal{H}_{BdG}$ is defined as:

$$\mathcal{H}_{BdG} = \begin{pmatrix} \text{Re}(S_q) & i\text{Im}(S_q) \\ -i\text{Im}(S_q) & -\text{Re}(S_q) \end{pmatrix}.$$

The eigenvalues of $\mathcal{H}_{\text{BdG}}$ (see Eq. (57)) are determined from the equation:

$$\det(\mathcal{H}_{\text{BdG}} - EI) = 0.$$

Expanding the determinant:

$$\begin{aligned} \det \begin{pmatrix} \text{Re}(S_q) - E_q & i\text{Im}(S_q) \\ -i\text{Im}(S_q) & -\text{Re}(S_q) - E_q \end{pmatrix} &= 0, \\ (\text{Re}(S_q) - E_q)(-\text{Re}(S_q) - E_q) - (+i\text{Im}(S_q))(-i\text{Im}(S_q)) &= 0 \\ -(\text{Re}(S_q)^2 + E_q \text{Re}(S_q) - E_q \text{Re}(S_q) - E_q^2) - \text{Im}(S_q)^2 &= 0 \\ E_q^2 &= \text{Re}(S_q)^2 + \text{Im}(S_q)^2 \\ E_q &= \pm \sqrt{\text{Re}(S_q)^2 + \text{Im}(S_q)^2} \\ E_q &= \pm |S_q|. \end{aligned} \tag{70}$$

The eigenvectors of $\mathcal{H}_{\text{BdG}}$ (see Eq. (57)) are given as:

$$\Psi = \begin{pmatrix} X_q \\ Y_q \end{pmatrix} = \begin{pmatrix} \cos \theta_q \\ -i \sin \theta_q \end{pmatrix},$$

where $\tan(2\theta_q) = -\frac{\text{Im}(S_q)}{\text{Re}(S_q)}$.¹⁵

Having now found the eigenvectors Ψ , the BdG transformation takes the form:

$$\begin{aligned} \begin{pmatrix} f_q \\ f_{-q}^\dagger \end{pmatrix} &= T \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix} \\ \begin{pmatrix} f_q \\ f_{-q}^\dagger \end{pmatrix} &= \begin{pmatrix} \cos \theta_q & -i \sin \theta_q \\ -i \sin \theta_q & \cos \theta_q \end{pmatrix} \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix}. \end{aligned} \tag{71}$$

When $2\theta_q = 0$ the BdG transformation becomes:

$$\begin{pmatrix} f_q \\ f_{-q}^\dagger \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix}.$$

The cmf (f_q) remain identical to the qsp (a_q). There is no mixing between f_q and their antiparticles $f_{-q}^\dagger$. The pairing term Δ_q is zero. The system behaves like a system of free fermions with no pairing terms.

When $2\theta_q = \frac{\pi}{2}$, we have maximum pairing, and the BdG transformation is:

$$\begin{pmatrix} f_q \\ f_{-q}^\dagger \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix}. \tag{72}$$

The cmf f_q no longer exist as independent states. They have fully mixed with their antiparticles $f_{-q}^\dagger$ and have become Bogoliubov qsp. The pairing Δ_q is maximal, and each f_q has been combined with an antiparticle $f_{-q}^\dagger$ to form a superconducting state.

¹⁵The proof of the relation $\tan(2\theta_q) = -\frac{\text{Im}(S_q)}{\text{Re}(S_q)}$ and the explanation of the form of X_q and Y_q are found in Appendix D.

The angle $2\theta_q$ controls how strong the pairing is:

$$\begin{aligned} 2\theta_q = 0, & \rightarrow \text{No mixing and zero pairing.} \\ 0 < 2\theta_q < \frac{\pi}{2}, & \rightarrow \text{Partial pairing, partial mixing.} \\ 2\theta_q = \frac{\pi}{2}, & \rightarrow \text{Maximum pairing, complete mixing.} \end{aligned}$$

The unitary transformation gives us:

$$T^\dagger \mathcal{H}_{BdG} T = \begin{pmatrix} |S_q| & 0 \\ 0 & -|S_q| \end{pmatrix}.$$

Applying the BdG transformation (see Eq. (71)) to Eq. (68) yields the final, diagonalized form of the Hamiltonian of the system in the zero-flux sector (the procedure is analogous to that followed to derive the Hamiltonian in Eq. (63)):

$$\boxed{\tilde{\mathcal{H}}_0 = \sum_q |S_q| (2\alpha_q^\dagger \alpha_q - 1)} \quad (73)$$

By varying the parameters J_x , J_y and J_z in Eq. (73), we compute the energy spectra corresponding to all possible phases of the system, as presented in Figs. 9, 10 and 11. These phases are summarized in the overall phase diagram of Fig. 12.

5 Phase diagram and energy spectrum $E(q_x, q_y)$

Summary

In Sec. 5, based on [1], [2], [23], [24], [25], [26] we first present the physical significance of Hamiltonian (73). We then analyze the various phases into which the system can transition, along with the corresponding energy spectrum, and finally present the phase diagram of the system.

5.1 Gapless phase

We start from the final diagonalized Hamiltonian, as presented in Eq. (73):

$$\tilde{\mathcal{H}}_0 = \sum_q |S_q| (2\alpha_q^\dagger \alpha_q - 1).$$

We define the ground state as the state in which all qsp are in the ground level $E_{g,s}$:

$$\alpha_q |\mathcal{M}_0\rangle = 0, \quad \forall q. \quad (74)$$

- $|\mathcal{M}_0\rangle$: The state of the cmf in the zero flux sector.

The ground state energy $E_{g,s}$ is defined as:

$$E_{g,s} = \langle \mathcal{M}_0 | \tilde{\mathcal{H}}_0 | \mathcal{M}_0 \rangle = - \sum_q |S_q|. \quad (75)$$

The excited states are given by $\alpha^\dagger |\mathcal{M}_0\rangle$, while the corresponding energy values E_c are:

$$E_c = 2 \sum_q |S_q| \alpha_q^\dagger \alpha_q = \begin{cases} 2|S_q|, & \text{if } \alpha_q^\dagger \alpha_q = 1 \rightarrow \exists \text{ excited quasiparticle,} \\ 0, & \text{if } \alpha_q^\dagger \alpha_q = 0 \rightarrow \nexists \text{ excited quasiparticle.} \end{cases}$$

The system enters a phase without an energy gap (Phase B), known as the gapless phase (gp), when the following condition is satisfied:

$$S_q = 0, \text{ for some } q. \quad (76)$$

Condition (5.3) holds for two cases¹⁶:

In the isotropic case, i.e. $J_x = J_y = J_z$, the system transitions to the gp, where two Dirac points (similar to those of graphene [27]) appear at positions:

$$Q = \pm \left(\frac{2\pi}{3}, \frac{-2\pi}{3} \right).$$

At the Dirac points $S_q = 0$, hence also $E_c = 0$, indicating that the energy gap at these points is zero. Near the Dirac points, the energy increases linearly according to the relation $E \propto |q - Q|$, forming conical surfaces around the minimum Q of the spectrum (see Fig. 9). These surfaces are known as Dirac cones. An analogous structure is observed for the negative energy values, down to the complete filling of the ground level $E_{g.s.}$ by qsp (Fermi sea). The negative Dirac cones are inverted and have the same point Q as their apex (see Fig. 9). In the gp, the cmf dominate, as they are the only gapless particles (residing in the ground level), in contrast to the cbf which exhibit an energy gap (gapped). At low temperatures, the cbf are arbitrarily excited to higher energy levels without energy cost. The excited cmf move freely within a fixed $\mathbb{Z}_2$ gauge field and behave as massless Dirac fermions (mdf).

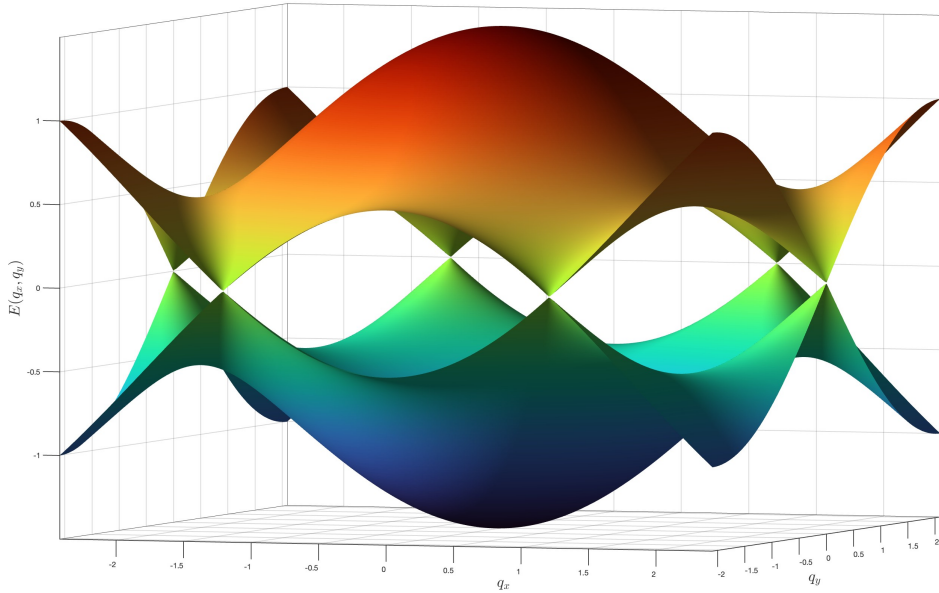


Figure 9: The energy spectrum $E(q_x, q_y)$ is shown for the isotropic case $J_x = J_y = J_z = 1$. Six zero-energy points (Dirac points) are observed within the 1st Brillouin Zone (BZ), where $E(q_x, q_y) = 0$, indicating that the spectrum is gapless. The dispersion near these points is linear, forming characteristic Dirac cones.

¹⁶The proof that condition (5.3) holds for the specific cases is presented in detail in Appendix E.

When all three triangle inequalities hold, the system is in the gp:

$$|J_z| < |J_x| + |J_y|, \quad (77)$$

$$|J_y| < |J_x| + |J_z|, \quad (78)$$

$$|J_x| < |J_z| + |J_y|. \quad (79)$$

As we vary the parameters J_x, J_y, J_z , the Dirac points shift and begin to merge. When even one of the inequalities (77), (78) or (79) ceases to hold, all Dirac points merge and eventually disappear. The energy spectrum acquires a gap and the system transitions to the phase with an energy gap, known as the gapped phase (ggp).

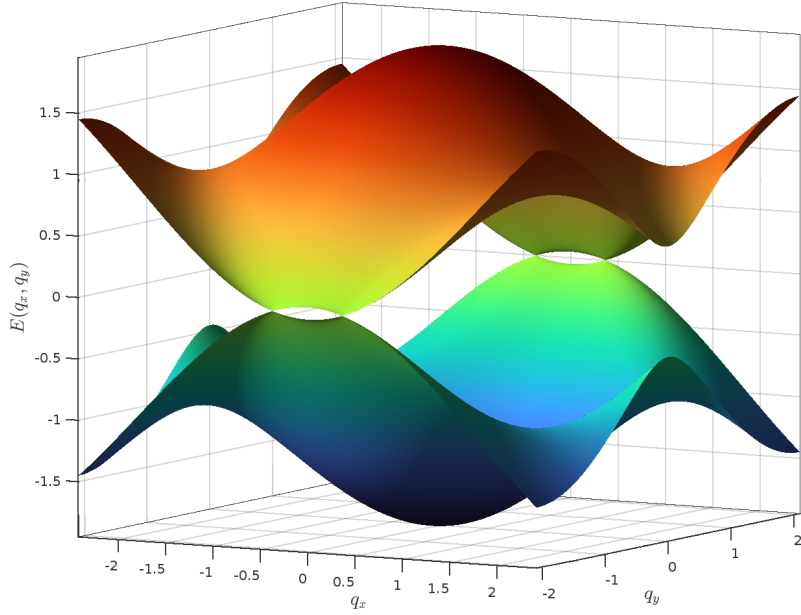


Figure 10: The energy spectrum $E(q_x, q_y)$ is shown for the boundary case $J_x = J_y = 1$ and $J_z = 1.9$. We observe that there are only four Dirac cones within the 1st BZ, where $E(q_x, q_y)$ vanishes. The Dirac points have begun to merge.

In the gp, the system behaves as a gapless quantum spin liquid [28] (QSL) with short-range correlations, due to the presence of Dirac points where the energy spectrum vanishes, allowing the existence of quantum excitations without an energy gap, which leads to strong quantum entanglement and fractionalization of the original spin degrees of freedom into two different types of fractional excitations¹⁷.

5.2 Gapped phase

The system transitions to the ggp when the condition:

$$S_q > 0, \text{ for all } q^{18} \quad (80)$$

is satisfied.

As mentioned in Section 5.1, if even one of the inequalities (77), (78) and (79) ceases to hold, the system acquires an energy gap and transitions to the ggp. Specifically, depending

¹⁷The proof for the characterization of the system as a QSL is presented in detail in Appendix F.

¹⁸The proof of why the system transitions to the ggp when condition (5.7) holds is presented in detail in Appendix E.

on which inequality is violated, the system transitions to phase A_{α_i} :

$$|J_z| > |J_x| + |J_y|, \rightarrow \text{Phase } A_z, \quad (81)$$

$$|J_y| > |J_x| + |J_z|, \rightarrow \text{Phase } A_y, \quad (82)$$

$$|J_x| > |J_z| + |J_y|, \rightarrow \text{Phase } A_x. \quad (83)$$

Depending on which phase A_{α_i} we are in, the specific α_i bonds dominate; e.g. in phase A_z the z bonds dominate in the system.

We assume we are in phase A_z .¹⁹

Once $|J_z|$ becomes larger than the sum $|J_x| + |J_y|$, the system enters phase A_z . The correlations in the z -bonds ($\mathcal{S}_{ij}^{zz}(t) = \langle \sigma_i^z(t) \sigma_j^z(0) \rangle$) begin to dominate. The correlations in the x and y bonds gradually weaken. When $J_z \gg J_x, J_y$, complete secondary coupled phase full dimerization appears in the z -bonds. The remaining x and y correlations are essentially zero (see Fig. F.2):

$$\begin{aligned} \langle \sigma_i^z(t) \sigma_j^z(0) \rangle &\approx 1, \\ \langle \sigma_i^x(t) \sigma_j^x(0) \rangle &\approx 0, \\ \langle \sigma_i^y(t) \sigma_j^y(0) \rangle &\approx 0. \end{aligned}$$

In phase A_{α_i} , the system breaks up into local dimers²⁰ along the bonds of type α_i , which have strong correlation, while the remaining directions are practically disconnected. When we say that the remaining directions are practically disconnected, we mean that the spins in the other two bonds remain frozen. They do not contribute to the ground state energy. The ground state of the system chooses to minimize its energy only through the dominant bonds (e.g. z -bonds in phase A_z). In phase A_{α_i} , the flux excitations (cbf) are at lower energy levels than those of the cmf, i.e. in contrast to the gp. The creation of a flux pair (e.g. two π -fluxes in one plaquette) has a lower energy cost compared to exciting a cmf. We again assume we are in phase A_z :

In the limit $J_z \gg J_x, J_y$, practically $J_x = J_y$, the cmf become very heavy, i.e. they have a large gap ($\Delta_{cmf} \propto J_z$). Fluxes can only be created via J_x, J_y , which are cheaper excitations:

$$\Delta_{flux} \propto \frac{J_x^4}{J_z^3}. \quad (84)$$

Eq. (84) shows that fluxes are activated only due to a weak perturbation in the dimers and confirms that phase A_z is deeply dimerized.

¹⁹Before proceeding, it is recommended to first read Appendix F, which is related to the correlation function $\mathcal{S}_{ij}^{\alpha\beta}(t)$.

²⁰A dimer is a pair of coupled particles (here spins), which are strongly correlated with each other (maximum correlation) and isolated from the rest of the system.

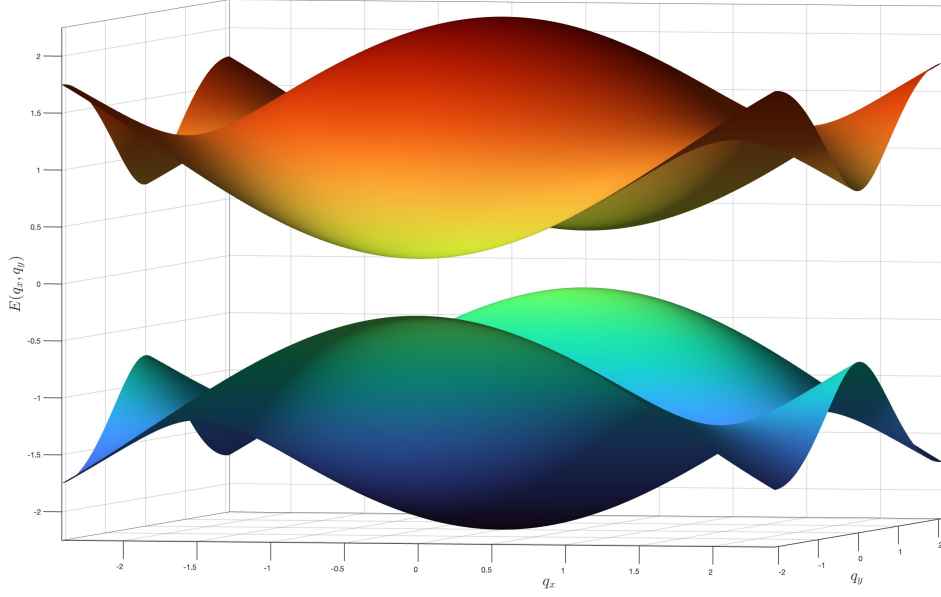


Figure 11: The spectrum $E(q_x, q_y)$ is shown above for $J_x = J_y = 1$ and $J_z = 2.5$. The system exhibits an energy gap and is in phase A_z .

In the ggp, the system behaves as a gapped QSL²¹ due to the appearance of an energy gap in the spectrum, while strong quantum entanglement and fractionalized excitations remain, preserving its disordered character.

5.3 Phase diagram

The phase diagram is triangular, satisfying the relation $J_x + J_z + J_y = 1$, and summarizes all the possible phases discussed in Sections 5.1 and 5.2.

²¹The proof of why the system is characterized as a QSL is presented in detail in Appendix F.

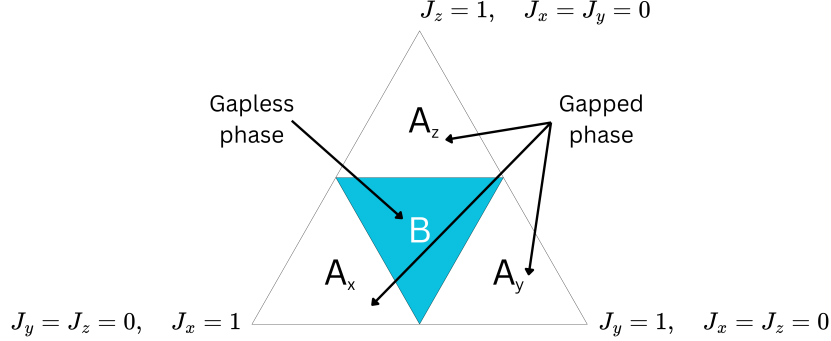


Figure 12: Phase diagram of the Kitaev model in the parameter space J_x, J_y, J_z . The regions A_{α_i} exhibit an energy gap (gapped), while region B is gapless. A point inside the above triangle describes the relative values of J_x, J_y, J_z . The isotropic point appears at the center of the triangle.

A Example of sectors in Hilbert space

Let us assume we have $n = 6$ (vertices), therefore:

$m = \frac{n}{2} = 3$ plaquettes

Dimension of the full Hilbert space: $2^n = 2^6 = 64$

Dimension of a sector: $2^{\frac{n}{2}} = 2^3 = 8$ (degrees of freedom)

Number of sectors: $64/8 = 8$ (as expected)

The full Hilbert space $\mathcal{L}$ consists of 8 sectors, and each sector will consist of a different combination of eigenvalues w_1, w_2, w_3 each time. All possible combinations are 8, equal to the number of sectors (as expected).

B Pauli Matrices

For particles with spin $\pm\frac{1}{2}$, the spinor space has dimension $n = 2$. The eigenvalue $+\frac{1}{2}$ corresponds to the quantum state $|\uparrow\rangle$, while the eigenvalue $-\frac{1}{2}$ corresponds to the state $|\downarrow\rangle$. We define the operator $\hat{S}_z$, which corresponds to the coupling of the spin along the z axis; its action on the eigenstates gives:

$$\hat{S}_z|\uparrow\rangle = \frac{\hbar}{2}|\uparrow\rangle \quad \text{and} \quad \hat{S}_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle. \quad (\text{B.1})$$

The representation of the operator $\hat{S}_z$ in the spinor space is given by the corresponding matrix:

$$\hat{S}_z = \begin{pmatrix} \langle\uparrow|\hat{S}_z|\uparrow\rangle & \langle\uparrow|\hat{S}_z|\downarrow\rangle \\ \langle\downarrow|\hat{S}_z|\uparrow\rangle & \langle\downarrow|\hat{S}_z|\downarrow\rangle \end{pmatrix} = \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

To compute the representation of the operators $\hat{S}_x$ and $\hat{S}_y$, we introduce the raising operator $\hat{S}_+$ and the lowering operator $\hat{S}_-$, so the operators $\hat{S}_x$ and $\hat{S}_y$ transform as:

$$\hat{S}_+ = \hat{S}_x + i\hat{S}_y, \quad \hat{S}_- = \hat{S}_x - i\hat{S}_y. \rightarrow \hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-), \quad \hat{S}_y = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-).$$

We know that the action of the operators $\hat{S}_+$ and $\hat{S}_-$ on the quantum states is:

$$\hat{S}_\pm |(s, m_s)\rangle = \hbar\sqrt{s(s+1) - m_s(m_s \pm 1)} |(s, m_s \pm 1)\rangle. \quad (\text{B.2})$$

Therefore the representation of the operator $\hat{S}_y$ in the spinor space is given by the corresponding matrix:

$$\hat{S}_y = \frac{\hbar}{2i} \begin{pmatrix} \langle \uparrow | \hat{S}_+ - \hat{S}_- | \uparrow \rangle & \langle \uparrow | \hat{S}_+ - \hat{S}_- | \downarrow \rangle \\ \langle \downarrow | \hat{S}_+ - \hat{S}_- | \uparrow \rangle & \langle \downarrow | \hat{S}_+ - \hat{S}_- | \downarrow \rangle \end{pmatrix} = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

The representation of the operator $\hat{S}_x$ in the spinor space is given by the corresponding matrix:

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} \langle \uparrow | \hat{S}_+ + \hat{S}_- | \uparrow \rangle & \langle \uparrow | \hat{S}_+ + \hat{S}_- | \downarrow \rangle \\ \langle \downarrow | \hat{S}_+ + \hat{S}_- | \uparrow \rangle & \langle \downarrow | \hat{S}_+ + \hat{S}_- | \downarrow \rangle \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We now define the operator $\vec{\hat{S}}$:

$$\vec{\hat{S}} = \frac{\hbar}{2} \vec{\hat{\sigma}}, \quad (\text{B.3})$$

where $\vec{\hat{\sigma}}$ are the Pauli matrices:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Properties of Pauli matrices:

Anticommutation relation:

$$\{\sigma^i, \sigma^j\} = 2\delta_{ij}I. \quad (\text{B.4})$$

Commutation relation:

$$[\sigma^i, \sigma^j] = 2i\epsilon_{ijk}\sigma^k. \quad (\text{B.5})$$

C Transformation of the flux operator $\hat{w}_{p_n}$

The initial representation of the plaquette on the left obeys the flux operator:

$$\hat{w}_{p_n} = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z.$$

Introducing the new formalism given by relation (1.3), through the operator K_{ij} , the plaquette transforms into the form shown on the right of the Figure. To understand how the flux operator $\hat{w}_{p_n}$ transforms, we first examine the external bonds. For example, particle number 1 participates in an x -bond, while particle 2 participates in a y -bond. The same procedure is followed for the rest. We define i as the particles with black dots and j as the particles with white dots. Combining the two steps we just saw together with relation (1.3), each dot takes the notation shown in blue in the Figure on the right. We are now ready to define the new formalism for the flux operator $\hat{w}_{p_n}$, which is defined as:

$$\begin{aligned} \hat{w}_{p_n} &= (\sigma_1^x \sigma_2^y)(\sigma_2^y \sigma_3^z)(\sigma_3^z \sigma_4^x)(\sigma_4^x \sigma_5^y)(\sigma_5^y \sigma_6^z)(\sigma_6^z \sigma_1^x) \\ &= K_{12}K_{23}K_{34}K_{45}K_{56}K_{61}, \quad \text{where } K_{ij} = K_{ji}. \end{aligned}$$

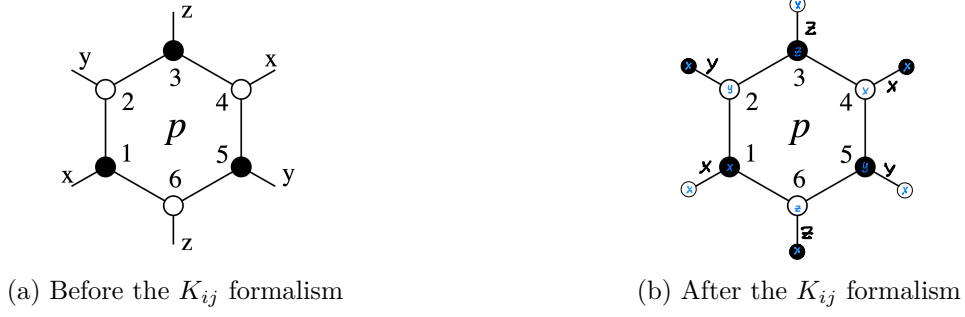


Figure 13: Comparison of a plaquette before and after the application of the K_{ij} formalism.

D Proof of relationship $\tan 2\theta_q = -\frac{Im(S_q)}{Re(S_q)}$

The BdG Hamiltonian is:

$$\mathcal{H}_{BdG} = \begin{pmatrix} h_q & -\Delta_q \\ -\Delta_q^* & -h_q \end{pmatrix}.$$

Eq. (4.9) is:

$$\mathcal{H}_{BdG}\Psi_q = E_q\Psi_q.$$

In matrix form eq. (4.9) takes the form:

$$\begin{pmatrix} h_q & -\Delta_q \\ -\Delta_q^* & -h_q \end{pmatrix} \begin{pmatrix} X_q \\ Y_q \end{pmatrix} = E_q \begin{pmatrix} X_q \\ Y_q \end{pmatrix}. \quad (85)$$

From eq. (D.1) we obtain the following equations:

$$h_q X_q - \Delta_q Y_q = E_q X_q, \quad (86)$$

$$-\Delta_q^* X_q - h_q Y_q = E_q Y_q. \quad (87)$$

From eq. (D.2) we find the ratio:

$$\frac{Y_q}{X_q} = \frac{E_q - h_q}{-\Delta_q}. \quad (88)$$

We have shown in Chap. 4:

$$\begin{aligned} E_q &= \pm \sqrt{Re(S_q)^2 + Im(S_q)^2}, \\ h_q &= Re(S_q), \\ \Delta_q &= -iIm(S_q). \end{aligned}$$

Substituting the above relations into eq. (D.4) leads to:

$$\begin{aligned} \frac{Y_q}{X_q} &= \frac{\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q)}{iIm(S_q)} \\ &= -i \frac{\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q)}{Im(S_q)}. \end{aligned}$$

We define the tangent $\tan \theta_q$ through the relation:

$$\tan \theta_q = \frac{\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q)}{Im(S_q)}. \quad (89)$$

Therefore

$$\frac{Y_q}{X_q} = -i \tan \theta_q. \quad (90)$$

From eq. (D.6) we prove that the eigenvectors of eq. (4.9) have the following form:

$$\Psi = \begin{pmatrix} X_q \\ Y_q \end{pmatrix} = \begin{pmatrix} \cos \theta_q \\ -i \sin \theta_q \end{pmatrix}.$$

Next we use the double angle identity:

$$\tan 2\theta_q = \frac{2 \tan \theta_q}{1 - \tan^2 \theta_q}. \quad (91)$$

Squaring both sides of eq. (D.5), we obtain:

$$\tan^2 \theta_q = \frac{(\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q))^2}{Im(S_q)^2}. \quad (92)$$

We expand the numerator:

$$\begin{aligned} (\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q))^2 &= Re(S_q)^2 + Im(S_q)^2 - 2Re(S_q)\sqrt{Re(S_q)^2 + Im(S_q)^2} + Re(S_q)^2 \\ &= 2Re(S_q)^2 + Im(S_q)^2 - 2Re(S_q)\sqrt{Re(S_q)^2 + Im(S_q)^2}. \end{aligned}$$

Substituting eq. (D.8) into the expansion of the numerator leads to:

$$\begin{aligned} \tan^2 \theta_q &= \frac{2Re(S_q)^2 + Im(S_q)^2 - 2Re(S_q)\sqrt{Re(S_q)^2 + Im(S_q)^2}}{Im(S_q)^2}, \\ 1 - \tan^2 \theta_q &= \frac{Im(S_q)^2 - (2Re(S_q)^2 + Im(S_q)^2 - 2Re(S_q)\sqrt{Re(S_q)^2 + Im(S_q)^2})}{Im(S_q)^2} \\ &= \frac{2Re(S_q)(\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q))}{Im(S_q)^2}. \end{aligned} \quad (93)$$

Substituting eq. (D.9) and (D.5) into eq. (D.7) leads to the final form:

$$\begin{aligned} \tan 2\theta_q &= \frac{\frac{2(\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q))}{Im(S_q)}}{\frac{2Re(S_q)(\sqrt{Re(S_q)^2 + Im(S_q)^2} - Re(S_q))}{Im(S_q)^2}} \\ &= \frac{1}{\frac{Re(S_q)}{Im(S_q)}} \\ &= -\frac{Im(S_q)}{Re(S_q)}, \quad \text{QED.} \end{aligned}$$

We have thus proved that:

$$\tan 2\theta_q = -\frac{Im(S_q)}{Re(S_q)}. \quad (94)$$

E Relationship between phase transitions and S_q

We start from eq. (4.21):

$$S_q = \sum_{n_i} J_{\alpha_{n_i}} e^{iqn_i}. \quad (95)$$

We expand eq. (E.1):

$$S_q = J_x e^{iqn_x} + J_y e^{iqn_y} + J_z. \quad (96)$$

Squaring both sides of eq. (E.2):

$$\begin{aligned} S_q^2 &= (J_x e^{-iqn_x} + J_y e^{-iqn_y} + J_z)(J_x e^{iqn_x} + J_y e^{iqn_y} + J_z) \\ &= J_x^2 + J_y^2 + J_z^2 + J_x J_z (e^{iqn_x} + e^{-iqn_x}) + J_y J_x (e^{iq(n_x - n_y)} + e^{-iq(n_x - n_y)}) + J_y J_z (e^{iqn_y} + e^{-iqn_y}). \end{aligned}$$

We use the identity $e^{i\theta} + e^{-i\theta} = 2\cos(\theta)$:

$$S_q^2 = J_x^2 + J_y^2 + J_z^2 + 2J_x J_z \cos(qn_x) + 2J_y J_x \cos(q(n_x - n_y)) + 2J_y J_z \cos(qn_y). \quad (97)$$

In order to be in the gapless phase (phase B), the condition $S_q = 0$ must hold for some value of q . Given that $-1 \leq \cos(x) \leq 1$, the minimum of the quantity S_q^2 is given by:

$$S_q^2 \geq J_x^2 + J_y^2 + J_z^2 - 2J_x J_z - 2J_y J_x - 2J_y J_z.$$

For the absolute minimum to be $S_q^2 = 0$, we need:

$$J_x^2 + J_y^2 + J_z^2 - 2J_x J_z - 2J_y J_x - 2J_y J_z \leq 0 \quad (98)$$

Rearranging the terms of eq. (E.4) leads to:

$$(J_x - J_y)^2 + (J_y - J_z)^2 + (J_z - J_x)^2 \leq 0 \quad (99)$$

Eq. (E.5) holds for two cases.

Case 1:

At the isotropic point:

$$J_x = J_y = J_z. \quad (100)$$

Case 2:

When all three triangle inequalities hold:

$$\begin{aligned} |J_z| &< |J_x| + |J_y|, \\ |J_y| &< |J_x| + |J_z|, \\ |J_x| &< |J_z| + |J_y|. \end{aligned} \quad (101)$$

When neither of the two eqs. (E.6) and (E.7) holds, then we have $S_q^2 > 0$ for all q and the system is now in the gapped phase (Phase A).

F Correlation function $\mathcal{S}_{\langle ij \rangle}^{\alpha\beta}(t)$

We first use eq. (3.5):

$$\begin{aligned} \sigma_j^{\alpha_i} &= \hat{D} \tilde{\sigma}_j^{\alpha_i}, \quad D = +1 \\ &= i \gamma_i^{\alpha_i} \gamma_j. \end{aligned}$$

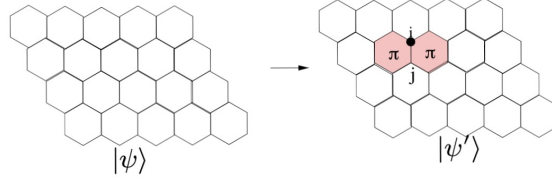


Figure 14: It is shown how a spin decomposes into two static π -fluxes and one dynamic cmf. The state $|\psi\rangle$ is a zero-flux state. Suppose the state $|\psi\rangle$ is in the zero-flux sector. Applying the operator σ_i^z to the state $|\psi\rangle$, where site i is connected to site j through the bond $\langle ij\rangle$, yields the new state $|\psi'\rangle$. The state $|\psi'\rangle$ contains two static π -fluxes in the plaquettes between the bond $\langle ij\rangle$, as well as one dynamic cmf, represented as a black circle.

We then express $\gamma_i^{\alpha_i}$ in the cbf basis (see eq. (3.14))
 $(\alpha \equiv \alpha_i)$

$$\sigma_j^{\alpha_i} = i\gamma_j(\chi_{\langle ij\rangle_\alpha} + \chi_{\langle ij\rangle_\alpha}^\dagger) \rightarrow i\gamma_j \hat{\pi}_{1_{\langle ij\rangle_\alpha}} \hat{\pi}_{2_{\langle ij\rangle_\alpha}}. \quad (102)$$

We define the operators $\hat{\pi}_{1_{\langle ij\rangle_\alpha}}$ and $\hat{\pi}_{2_{\langle ij\rangle_\alpha}}$ whose action adds a π flux to plaquette 1 and 2 located between the bond $\langle ij\rangle_\alpha$ (see Fig. F.1).

Based on eq. (F.1) and Fig. (F.1), the action of the spin operator σ_j^α on the state $|\psi\rangle = |\mathcal{G}\rangle |\mathcal{M}_\mathcal{G}\rangle$ is defined as:

$$\sigma_j^\alpha |\mathcal{G}\rangle |\mathcal{M}_\mathcal{G}\rangle = \gamma_j |\mathcal{G}^{i\alpha}\rangle |\mathcal{M}_\mathcal{G}\rangle. \quad (103)$$

- $|\mathcal{G}\rangle$: State of the $\mathbb{Z}_2$ field.
- $|\mathcal{M}_\mathcal{G}\rangle$: State of the cmf in the specific sector $\mathcal{G}$.
- $|\mathcal{G}^{i\alpha}\rangle$: New $\mathbb{Z}_2$ field state where we have changed the local gauge field configuration at site i .

The magnetization of the system is defined as $\mathcal{M}_\alpha = \frac{1}{N} \langle \sigma_i^\alpha \rangle$:

$$\begin{aligned} \langle \sigma_i^\alpha \rangle &= \langle \mathcal{M}_\mathcal{G} | \langle \mathcal{G} | \sigma_i^\alpha | \mathcal{G} \rangle | \mathcal{M}_\mathcal{G} \rangle \\ &= \langle \mathcal{M}_\mathcal{G} | \langle \mathcal{G} | \gamma_i |\mathcal{G}^{i\alpha}\rangle | \mathcal{M}_\mathcal{G} \rangle \\ &= \gamma_i \langle \mathcal{M}_\mathcal{G} | \mathcal{M}_\mathcal{G} \rangle \langle \mathcal{G} | \mathcal{G}^{i\alpha} \rangle, \quad \langle \mathcal{G} | \mathcal{G}' \rangle = \delta_{n_\mathcal{G}, n_{\mathcal{G}'}} \\ &= 0, \end{aligned}$$

where $n_\mathcal{G}$ corresponds to the distribution of cbf in the specific flux sector.

We have therefore shown that the magnetization in our system is zero, i.e., $\mathcal{M}_\alpha = 0$. The dynamic correlation function (dcf) of the system is defined as:

$$\mathcal{S}_{ij}^{\alpha\beta}(t),$$

where the index α refers to directions corresponding to nearest-neighbor bonds ($n = 1$), while the index β concerns higher-order neighbors $n > 1$.

²²It is sometimes confusing that, after the action of the spin operator in eq. (F.2), the state $|\mathcal{M}_\mathcal{G}\rangle$ appears unchanged, as the notation remains the same. However, this is done exclusively for symbolic simplification and does not mean that the state remains invariant.

$$\mathcal{S}_{\langle ij \rangle}^{\alpha\beta}(t) = \langle \sigma_i^\alpha(t) \sigma_j^\beta(0) \rangle = \langle \mathcal{M}_{\mathcal{G}} | \langle \mathcal{G} | \sigma_i^\alpha(t) \sigma_j^\beta(0) | \mathcal{G} \rangle | \mathcal{M}_{\mathcal{G}} \rangle. \quad (104)$$

We use the Heisenberg representation: $A(t) = e^{iHt} \hat{A} e^{-iHt}$.

Eq. (F.3) describes the probability of finding a spin at site j (for $t=0$) with coordinates β and another spin at site i (after time t) with coordinates α .

$$\begin{aligned} \mathcal{S}_{\langle ij \rangle}^{\alpha\beta}(t) &= \langle \mathcal{M}_{\mathcal{G}} | \langle \mathcal{G} | \sigma_i^\alpha(t) \sigma_j^\beta(0) | \mathcal{G} \rangle | \mathcal{M}_{\mathcal{G}} \rangle \\ &= \langle \mathcal{M}_{\mathcal{G}} | \langle \mathcal{G} | \sigma_i^\alpha(t) \gamma_j(0) | \mathcal{G}^{i\beta} \rangle | \mathcal{M}_{\mathcal{G}} \rangle \\ &= \gamma_j(0) \langle \mathcal{M}_{\mathcal{G}} | \langle \mathcal{G} | \sigma_i^\alpha(t) | \mathcal{G}^{i\beta} \rangle | \mathcal{M}_{\mathcal{G}} \rangle \\ &= \gamma_j(0) \gamma_i(0) e^{i(H-E)t} \langle \mathcal{M}_{\mathcal{G}} | \mathcal{M}_{\mathcal{G}} \rangle \langle \mathcal{G} | \mathcal{G}^{i\alpha} \rangle \\ &= \gamma_j(0) \gamma_i(0) e^{i(H-E)t} \delta_{n_{\mathcal{G}}, n_{\mathcal{G}^{i\alpha}}} \\ \mathcal{S}_{\langle ij \rangle}^{\alpha\beta}(t) &= \begin{cases} g_{\langle ij \rangle \alpha}(t) \delta_{\alpha, \beta}, & \text{if } i \text{ and } j \text{ are nearest neighbors,} \\ 0, & \text{anything else.} \end{cases} \end{aligned} \quad (105)$$

Based on eq. (F.4) we showed that the dcf vanishes beyond nearest neighbors.

For simplification of the operations we continue our analysis for $t = 0$. We compute $\mathcal{S}_{\langle ij \rangle}^{\alpha\alpha}(0)$ based on eq. (F.4):

$$g_{\langle ij \rangle \alpha}(0) = \mathcal{S}_{\langle ij \rangle}^{\alpha\alpha}(0) = \langle \mathcal{M}_{\mathcal{G}} | \langle \mathcal{G} | \gamma_j(0) \gamma_i(0) | \mathcal{G} \rangle | \mathcal{M}_{\mathcal{G}} \rangle. \quad (106)$$

Applying a Fourier transform to the cmf, eq. (F.5) within the 1st BZ transforms into integral form:

$$\gamma_j(0) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}_1} \gamma_{\mathbf{q}_1}(0) e^{i\mathbf{q}_1 \cdot \mathbf{n}_x}, \quad \gamma_i(0) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}_2} \gamma_{\mathbf{q}_2}^\dagger(0) e^{-i\mathbf{q}_2 \cdot \mathbf{n}_y}.$$

$$\mathcal{S}_{\langle ij \rangle}^{\alpha\alpha}(0) = \frac{\sqrt{3}}{16\pi^2} \int_{BZ} \langle \gamma_{\mathbf{q}_1} | \gamma_{\mathbf{q}_2}^\dagger \rangle e^{i\mathbf{q} \cdot (\mathbf{n}_x - \mathbf{n}_y)} dq_1 dq_2. \quad (107)$$

From the completeness relation we know that $\langle \gamma_{\mathbf{q}_1} | \gamma_{\mathbf{q}_2}^\dagger \rangle = \frac{h_q}{E_q}$. Moreover, we assume that $e^{ik(n_1 - n_2)} \approx 1$, since we restrict to nearest neighbors (very short distances).

The energy of the quasiparticles is given by:

$$E_q = \sqrt{h_q^2 + \Delta_q^2} \quad (\text{see Chap. 4}).$$

We define the angular functions $\theta(q_1, q_2)$ as:

$$\cos \theta(q_1, q_2) = \frac{h_q}{E_q}, \quad \sin \theta(q_1, q_2) = \frac{\Delta_q}{E_q},$$

where:

$$q_1 = \mathbf{q} \cdot \mathbf{n}_x, \quad q_2 = \mathbf{q} \cdot \mathbf{n}_y.$$

The final form of the dcf $\mathcal{S}_{\langle ij \rangle}^{\alpha\alpha}(0)$ (see eq. (F.6)) now reads:

$$\mathcal{S}_{\langle ij \rangle}^{\alpha\alpha}(0) = \frac{\sqrt{3}}{16\pi^2} \int_{BZ} \cos \theta(q_1, q_2) dq_1 dq_2. \quad (108)$$

Using the Monte Carlo method, we numerically estimate eq. (F.7) for $\alpha = z$ and compute the dcf for coupling constants $J_x = J_y = 0.5$, varying the value of J_z in the interval $[0, 1]$. The corresponding diagram of the dcf has been taken directly from [23],

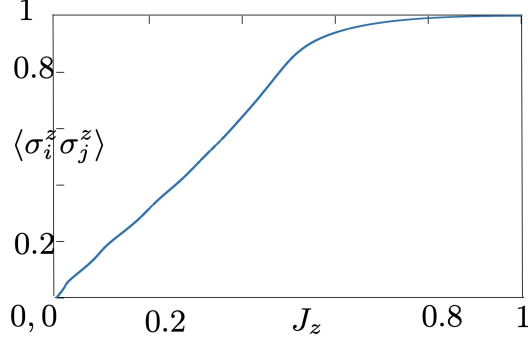


Figure 15: The above graph shows the correlation $\langle \sigma_{1z} \sigma_{2z} \rangle$ on a given z-bond. We have set $J_x = J_y = 0.5$ and vary J_z from 0 to 1. As expected, for $J_z = 0$ the correlation is zero, while for large values of J_z it converges to 1.

The correlation function $\mathcal{S}_{\langle ij \rangle}^{\alpha\beta}(0)$ is zero for distances beyond nearest neighbors, i.e.:

$$\mathcal{S}_{\langle ij \rangle}^{\alpha\beta}(0) = 0, \quad \text{for } \langle ij \rangle_n \text{ neighbors where } n > 1.$$

While for nearest neighbors, the correlation is not simply constant, but follows a characteristic power-law behavior:

$$\mathcal{S}_{\langle ij \rangle}^{\alpha\alpha}(0) \sim \frac{1}{|i - j|^n}.$$

This behavior is a clear indication that the system is in a Quantum Spin Liquid (QSL) phase, specifically a short-range liquid phase without macroscopic order.

G MATLAB Implementation for Computing the Energy Spectrum $E(q_x, q_y)$

The following code computes the energy spectrum $E(q_x, q_y)$ of the Kitaev model on a honeycomb lattice. This particular example corresponds to the parameter set $(J_x = 1, J_y = 1, J_z = 2.5)$, which describes the Abelian phase A_z . By appropriately modifying the coupling constants, all energy spectra corresponding to the various phases of the system can be generated.

```
% Couplings
Jx = 1;
Jy = 1;
Jz = 2.5;
N = 300;

% Brillouin zone sampling
qx_range = linspace(-4*pi/(3*sqrt(3)), 4*pi/(3*sqrt(3)), N);
qy_range = linspace(-2*pi/3, 2*pi/3, N);
[qx, qy] = meshgrid(qx_range, qy_range);
```

```

% Dispersion for anisotropic couplings
term1 = cos(sqrt(3)*qx);
term2 = cos( (3*qy)/2 - (sqrt(3)*qx)/2 );
term3 = cos( (3*qy)/2 + (sqrt(3)*qx)/2 );

Sq = 0.5 * sqrt( ...
    Jx^2 + Jy^2 + Jz^2 + ...
    2*Jx*Jy*term1 + ...
    2*Jy*Jz*term2 + ...
    2*Jz*Jx*term3 );

% Hexagonal mask
mask = abs(qx * sqrt(3)/2) + abs(qy * 3/2) <= 2*pi;
Sq(~mask) = NaN;

% Plot
figure;

surf(qx, qy, Sq, ...
    'FaceColor', 'interp', ...
    'EdgeColor', 'k', ...
    'LineStyle', '-', ...
    'LineWidth', 0.3); hold on;

surf(qx, qy, -Sq, ...
    'FaceColor', 'interp', ...
    'EdgeColor', 'k', ...
    'LineStyle', '-', ...
    'LineWidth', 0.3);

% Style
colormap turbo;
shading interp;
material dull;
lighting gouraud;
camlight headlight;

xlabel('$q_x$', 'Interpreter', 'latex', 'FontSize', 14);
ylabel('$q_y$', 'Interpreter', 'latex', 'FontSize', 14);
zlabel('$E(q_x, q_y)$', 'Interpreter', 'latex', 'FontSize', 14);
title('');

view(45, 25);
axis tight;
box on;
grid on;

% Grid style
ax = gca;
ax.LineWidth = 1;
ax.GridAlpha = 0.2;
ax.XColor = [0.2 0.2 0.2];
ax.YColor = [0.2 0.2 0.2];
ax.ZColor = [0.2 0.2 0.2];

```

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